

FTK SUITE 8.2 - ARTIFACTS GUIDE

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About Exterro

Exterro was founded with the simple vision that applying the concepts of process optimization and data science to how companies manage digital information and respond to litigation would drive more successful outcomes at a lower cost. We remain committed to this vision today. We deliver a fully integrated Data Risk Management platform that enables our clients to address their privacy, regulatory, compliance, digital forensics, and litigation risks more effectively and at lower costs. We provide software solutions that help some of the world's largest organizations, law enforcement and government agencies work smarter, more efficiently, and support the Rule of Law.

1 Supported Chat Apps

Application	Android	iOS
Android Messages	13.0.10.73	<i>Not Applicable</i>
Discord	176.21	176
Facebook messenger	406.0.0.13.115	412
Google Meet/chat	177.0.485451249.duo.android_20221002.73_p3.s	6.0.230417
IMOPlus	2023.04.2051	2023.4.5
Instagram	281.0.0.19.105	281
iOS Messages (iOS)	<i>Not Applicable</i>	iOS 14.0
KakaoTalk	10.1.8	10.1.8
Kik	15.51.1.28280	16.9.3
Line	13.6.1	13.6.1
LinkedIn (Chats)	4.1.818.1	9.1.318
Microsoft Teams (Chats)	1416/1.0.0.2023072702	5.8.0
Signal	6.19.6	0.417
Skype	8.96.0.409	Supported
Slack	23.04.40.0	Supported
Snapchat	12.33.0.19	Supported

Application	Android	iOS
Telegram	9.6.5	9.6.3
Threema	5.0.5	Supported
TikTok	<i>Not Applicable</i>	<i>Not Applicable</i>
Tinder (Partial Support)	13.21.1	Supported
Twitter (Chats, Posts, and Tweets)	9.87.0-release.0	9.56.1
Uber (Cached Data)	4.472.10002	3.569.10000
Viber	19.9.4.0	19.9.1
WeChat (Partial Support)	<i>Not Applicable</i>	<i>Not Applicable</i>
WhatsApp	2.23.8.76	23.8.78
Wickrme	6.2.7	6.226.4.2
Mega	11.0 (233270828) (odc1a146b7)	13.1.1
Burner	5.2.8	<i>Not Applicable</i>

Operating System Support

- **Android** – Android 7.0 Nougat to Android 15
- **iOS** – iOS 12 to iOS 18

File Types

- **Android:** Physical Images, File system images, android backup (adb) encrypted/unencrypted, UFD.
- **iOS:** Full File System Images, iOS backups, and File System images.

Mobile Chat Features

You can now view message replies from the following chat applications:

Message Replies Supported:

- Microsoft Teams
- WhatsApp
- Skype
- Line
- KakaoTalk
- iMO Plus
- Instagram
- Facebook Messenger
- Threema
- Viber

You can also view information regarding deleted or modified messages from the following chat applications:

Deleted Messages:

- iMO Plus
- Skype
- LinkedIn
- Google Chat
- WhatsApp
- Facebook Messenger
- Microsoft Teams
- KIK

Edited Messages:

- WhatsApp
- Viber
- Telegram
- Microsoft Teams
- Google Chat
- Skype
- Discord
- LinkedIn
- Slack
- KIK

Additional Features:

- **LinkedIn Drafts:** Support for LinkedIn chat drafts.
- **WhatsApp Polls:** Ability to view WhatsApp polls.
- **Facebook Secret Messages:** You can now view Facebook secret messages.

2 Chat Apps supported with Logical Collection

- Skype
- Kik
- Viber
- IMO
- WhatsApp
- Line
- Threema
- KakaoTalk
- Microsoft Teams (upto 7.5.0)
- Mega
- Tinder
- iMessage

3 Windows OS (Basic Layout)

Applications

Description	This section will contain information relating to applications which can be discovered and recovered using the Forensic Tool Kit. This includes application information such as install states, start up and auto run states; it also includes information regarding drivers. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, application name, file path, and file sizes may be found under the Installed attribute.
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Attribute	Associated Columns	
Installed	- User Name - File Path - Version - Publisher	- Application Name - Install Date - Size - Provenance
Prefetch	- User Name - Run Count - Application Path - File Created Date/Time 3rd Last Run Date/Time 5th Last Run Date/Time 7th Last Run Date/Time - Provenance	- Application Name - Last Run Time - Volume 2nd Last Run Date/Time 4th Last Run Date/Time 6thLast Run Date/Time 8th Last Run Date/Time
User Assist	- User Name - Run Count - File Path - Focus Seconds	- File Name - Last Run Time - Focus Count - Provenance

Attribute	Associated Columns	
AmCache Driver Binaries	• Driver Name • Last Installed Date/Time • Product • Service • Driver in Box • Signed • Company • Driver Timestamp • Key • Provenance	• Driver Version • Last Write Date/Time • Product Version • Key • Kernel Mode • Checksum • Driver ID • Driver Type • Image Size
AmCache Driver Packages	• Key • Class • Driver In Box • Inf • Submission ID • Provenance	• Last Installed Date/Time • Directory • HWIDs • Company • Sysfile
AmCache File Entries	• Name • File Extension • SHA1 Hash • Link Date • Size • Program ID • PE File • Bin Product Version • Associated Application Name	• Last Updated Date/Time • Key • OS Component • Product Name • Version • Key • Bin File Version • Language • Provenance

Attribute	Associated Columns	
AmCache PnP Devices	• Driver Name • Description • Class Value • Container ID • Driver Package Strong Name • Driver Version • HWID • Install Date • Matching ID • Parent ID • Provider • STACKID	• Last Updated Date/Time • Key • Class GUID • Driver ID • Driver Date • Enumerator • Inf • Manufacturer • Model • Problem Code • Service • Provenance
AmCache Program Entries	• Name • Install Date • Publisher • Inbox Modern App • Provenance	• Last Updated Date/Time • Version • Hidden App • File Path
AutoRun Items	• Name • Trigger Condition • Command • Provenance	• Type • File Path • Modified Date/Time
StartUp Items	• Name • Last Modified Date/Time • Deleted	• Type • Path • Provenance
System Services	• Name • Start Type • Dependencies	• Type • User Account • Description

Attribute	Associated Columns
	• Service Location • Display Name • Hosted Service • Error Control • Group • Hosted • Hosted Service Parameters • Provenance
Shim Cache	• Name • Last Updated Date/Time • File Path • Last Run Date/Time • Executed • Provenance

Device Interaction

Description	This section will contain information relating to device interactions which can be discovered and recovered using the Forensic Tool Kit. This includes information such as USB connections. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found.
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Attribute	Associated Columns	
USB Devices	- User Name - Last Removal Time - Vendor ID - Product ID - Revision - Last User Mount Time - Mount Locations - First Connect Since Reboot - Data Provenance	- Last Connect Time - Vendor - Product - Instance ID - Revision ID - First Connect Time (Local) - Volume Label - Friendly Name - Provenance
USB Volume Storage	- User Name - Instance ID - Last Removal - First Connect	- Last Connect - Provenance - Data Provenance
AmCache Device Containers	- User Name - Model - Device Type - Model ID - Discover Method - Active - Machine Container - Paired	- Device Friendly Name - Last Updated - Manufacturer - State - Icon - Connected - Networked

Browsers

Description	This section will contain information relating to browser information items which can be discovered and recovered using the Forensic Tool Kit. This includes application information such as downloads, searches, and history; also found here is information concerning categories of URLs. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, if looking at the Phishing URLs attribute, present nodes would include URL Title and Last Visit Time.
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Attribute	Associated Columns	
Keyword Searches	- User Name - Search Term - Web Page Title - Search Engine	- Date/Time - URL - Browser - Provenance
Credentials	- User - URL - Password	- User Name - Browser
Downloads	- User Name - URL - File Size	- File Path - Download Time - Browser
Cookies	- User Name - Value - Creation Time - Domain - Expiration Time	- Name - Last Access Time - Browser - Path
Rebuilt webpages	- Title - Created Date/Time - Cache Table	- URL - Domain
Social Media URLs	Name	Last Visit Time

Attribute	Associated Columns	
	• URL • Category • Description • Metadata	• URL Title • Sub Category • Activity
Darknet URLs	• Name • URL • Category • Description • Metadata	• Last Visit Time • URL Title • Sub Category • Activity
Cryptocurrency URLs	• Name • URL • Category • Description • Metadata	• Last Visit Time • URL Title • Sub Category • Activity
Tax Site URLs	• Name • URL • Category • Description • Metadata	• Last Visit Time • URL Title • Sub Category • Activity
Google Searches	• Search Term • URL • Date/Time • Page Load Date/Time • Activity	• Original Search Query • Web Page Title • Search Session Start Date/Time • Session Type
Google MAPS Queries	• Search Term • Source • Centre of Map • Activity	• Date/Time • Destination • Browser

Attribute	Associated Columns	
Phishing URLs	• Name • URL • Category • Description • Metadata	• Last Visit Time • URL Title • Sub Category • Activity
Pornography URLs	• Name • URL • Category • Description • Metadata	• Last Visit Time • URL Title • Sub Category • Activity

Networks

Description	This section will contain network information which can be discovered and recovered using the Forensic Tool Kit. This includes information such as network connections and wireless profiles. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, MAC addresses and IP addresses, as well as DNS and DHCP servers may be found under the Network Interfaces attribute.
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Attribute	Associated Columns	
Network Shares	- User Name - Last Connect Time	Network Path
Network Connections	- User Name - Last Connect Time - Profile Name - Connection Mode - Encryption	- First Connect Time - Network Name - Network Category - Authentication - Password
Wireless Profiles	- User Name - Wireless Authentication - Profile Name - Authentication - Password	- SSID - Wireless Encryption - Connection Mode - Encryption
Network Interfaces	- Adapter Name - Gateway MAC Address - DHCP Enabled - DHCP IPv4 Subnet Mask - DHCP Server - DHCP DNS Domain - Lease Obtained Date/Time - IPV4 Address - IPV4 Gateways - DNS Domain	- MAC Address - Is Wireless Connection - DHCP IPv4 Address - DHCP IPv4 Gateway - DHCP DNS Servers - DHCP IPv4 Address - Less Expiry Date/Time - IPV4 Subnet Mask - DNS Servers

Recent Files/Folders

Description	This section will contain information relating to recently used files and folders which can be discovered and recovered using the Forensic Tool Kit.
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Attribute	Associated Columns	
Keyword Searches	• User Name • Search Date/Time	• Search Term • Provenance
Shortcuts (LNK)	• User Name • Extension • Target Last Write Time • Target Creation Time • Volume Serial Number • Net Bios Name • MAC Address • Provenance • LF Create Date • LF Access Data	• File Name • Target File Path • Target Last Write • Target Attributes • Volume Name • Target File Size • Pinned To List • Arguments • LF Modify Data
Jump List	• User Name • Application Name • Target File Path • Run Count • Target Last Access • Target Last Write • Target Creation Time • Pinned to List • Application ID • Vol Name • Vol Serial# • Target NetBIOS Name	• Target MAC Address • Target File Size • Destlist Last Modified FT • Path • Category • Created • Drive Type • Arguments • Jump List Type • Provenance • Data Provenance

Attribute	Associated Columns	
AmCache Shortcuts	• User Name • Target Last Write Time • Path • Created	• Name • Shortcut Path • Category • Provenance
MRU Run Commands	• User Name • Target Last Write Time • Value Name	• Run Command • Registry Order • Provenance
MRU Recent Files & Folders	• User Name • File/Folder Link • Registry Order • Path • Created	• File/Folder Name • Target Last Write Time • Value • Category • Provenance
MRU Recent Open/Saved Files	• User Name • File/Folder Link • Registry Order • Path • Created	• File Name • Target Last Write Time • Value • Category • Provenance
MRU Folder Access	• User Name • Directory • Registry Order • Path • Created	• Application Name • Target Last Write Time • Value • Category • Provenance
Shell Bags	• User Name • Target Last Write Time • Target Creation Time • Bag Slot • Created	• Target File Path • Target Last Access Time • Type • Provenance • Registry Key Last Write Time

Attribute	Associated Columns
	• 1st Interacted • Reg Key • Category • Last Interacted • Path

Operating System Information

Description	This section will contain information relating to operating system information which can be discovered and recovered using the Forensic Tool Kit. This includes features such as owner and user account information. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, username and logon times may be found under the User Accounts attribute.
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Attribute	Associated Columns	
Desktop Items	• Usr Acct • Provenance	• Data Provenance
MUICache Information	• Usr ID • Usr Acct • Data	• Path • Provenance • Data Provenance
Shell Bags	• Bag Slot • Registry Key Last Written • 1st Interacted • Category • Created	• Last Interacted • Reg Key • Path • Provenance • Data Provenance
Owner Information	• Product Name • Current Version • Current Type • Product Key • DigitalProductID4 • Path • Software Type • Installation Type • EditionSubString • Current Version • CurrentMinorVersionNumber	• Edition ID • Install Date • Registered Owner • Product ID • DigitalProductID • System Root • ReleaseID • EditionSubVersion • Edition SubManufacturer • Current Type • Current Build Number

Attribute	Associated Columns	
	- Current Build - BuildLabEx - Build GUID	- CompositionEditionID - BuildLab - BuildBranch
Timezone Information	- Standard Timezone Name - Daylight Timezone Name - Standard Timezone Start Date/Time - Local Time	- Current Timezone Offset (Minutes) - Daylight Timezone offset (minutes) - Daylight Timezone Start Date/Time - Local Time
User Accounts	- User Name - SID - Last Login Date/Time	- User Type - Profile Path
Group Accounts	- Group Name - SID - Description - Rest of the user account columns	- Group Type - RID - Users in Group
Windows Notifications Center	- Notification - App Name - Received Date/Time - Created Date/Time - Provenance	- SubText - Notification Type - Expiry Date/Time - Modified Date/Time
Windows Timeline Activity	- Display Name - Content - End Date/Time - Focus (Seconds) - Platform - Last Modified Date/Time - OnClient Modification Date/Time	- Application Name - Start Date/Time - Type - Activity ID - Created Date/Time - Cloud Creation Date/Time - Provenance

SRUM

Description	This section will contain information relating to Windows System Resource Usage Monitor (SRUM) which can be discovered and recovered using the Forensic Tool Kit. SRUM collects statistics on the execution of binaries. This includes features such as network connections and usage. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, examining network usage by an application, the nodes under Storage Events attribute would provide information such as the application name, bytes sent and received, and the recorded dates.
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Attribute	Associated Columns	
SRUM Network Connections	- SSID - Duration - Recorded Date/Time	- Connection Start Date/Time - Interface Type - Provenance
SRUM Network Usage	- App Name - SSID - Bytes Received - Recorded Date/Time - Provenance	- SID - Bytes Sent - Interface Type - App Path
SRUM Push Notifications	- App Name - Recorded Date/Time - Network Type - Provenance	- SID - Notification Type - Payload Size

Windows Events

Description	This section will contain information relating to Windows Event Logs (EVTX) which can be discovered and recovered using the Forensic Tool Kit. EVTX files store system log information. This includes features such as Shell, Storage, Login/Logout and other events. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, examining network usage by an application, the nodes under Storage Events attribute would provide information such as the manufacturer, drive GUID, and serial numbers.
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Attribute	Associated Columns	
Account Management Events	- Event ID - Event Src - Created - Path	- Task Category - Event Level - Description - Category
Audit Policy Change Events	- Event Src - Created - Category - Data Provenance	- Event Level - Path - Provenance
Firewall Events	- Event Src - Event Data - Created - Provenance	- Event Level - Category - Path - Data Provenance
Kernel Events	- Event Src - Event Data - Created - Provenance	- Event Level - Category - Path - Data Provenance
Login/Logout Events	- Event ID - Event Src	- Task Category - Event Level

Attribute	Associated Columns	
	- Created - Path	- Description - Category
Network Auditing Events	- Event Src - Event Level - Category - Path - Data Provenance	- Event ID - Event Data - Created - Provenance
Other Windows Events	- Event Src - Event Data - Created - Provenance	- Event Level - Category - Path - Data Provenance
Power Entry Events	- Event Src - Event Data - Created - Provenance	- Event Level - Category - Path - Data Provenance
Remote Desktop Protocol Events	- Event Src - Event Data - Created - Provenance	- Event Level - Category - Path - Data Provenance
Scheduled Task Events	- Event Src - Category - Path - Data Provenance	- Event Level - Created - Provenance - Task Category
Security Events	- Event Src - Event Level - Category - Path	- Event ID - Event Data - Created - Provenance

Attribute	Associated Columns	
	Data Provenance	
Shared Object Events	- Event ID - Event Src - Created - Path	- Task Category - Event Level - Description - Category
Shell Events	- Event Src - Event Level - Category - Path - Data Provenance	- Event ID - Event Data - Created - Provenance
Storage Events	- Event Src - Event Level - Category - Vol Serial # - Disk Capacity - Registry Entry ID - Parent ID - Created - Version - Container ID - Task Count - Volume Correlation ID - Computer Name - Process ID - Provenance	- Event ID - Event Data - Model - Full Vol Serial # - Location Title - Manufacturer - Partition Style - Path - Drive Name - Work Time (ms) - Drive GUID - Vol ID - User SID - Thread ID - Data Provenance

4 Mac OS (Mac Column Sets)

Applications

Description	This section will contain information relating to applications which can be discovered and recovered using the Forensic Tool Kit. This includes application information such as install states and application permissions. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, application name, file path, and file sizes may be found under the Installed attribute.
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Attribute	Associated Columns
Installed	Apps-Installed (Mac) - User Name - Pkg Name - Installed - Internal Vers - App Store Action - Path - Created - Data Provenance - Application Name - File Path - Display Vers - File Size - Publisher - Category - Provenance
	Apps-Permission (Mac) - Application Name - Permission Granted - Created - Data Provenance - Service - Path - Provenance
Application Permissions - MAC	
Resumed Apps - MAC	Apps-Resumed Apps (Mac) - Application Name - Owning BundleID - Path - User Name - Bckgnd State - Created

Attribute	Associated Columns
	- Category Provenance - Data Provenance
StartUp Items - MAC	Apps- Startup Items (Mac) - Application Name - Startup Arg - Type - Proc Type - Disabled? - Path - Category - Created - Category Provenance - Data Provenance
Menu Bar Apps - MAC	Apps-Menu bar Items (Mac) - Application Name - Path - Category - Path - Created - Provenance - Data Provenance
Dock Items - MAC	Apps- Dock Items (Mac) - Pkg Name - User Name - State - Fldr Path - GUID - Path - Created - Provenance - Data Provenance
Core Analytics - MAC	Apps-Core_Analytics - Bundle ID - Active Time in Seconds #Activations #Launches - Proc Name - Ran if Fg - Uptime (s) - Version - Start Time - End Date - Category - Path - Created - Provenance - Data Provenance

Attribute	Associated Columns
KnowledgeC Application Activities	Apps- KnowledgeC Apps Activities • Bundle ID • Assoc URL Title • End Date • Category • Created • Data Provenance • Content Desc • Start Time • Entry Creation • Path • Provenance
KnowledgeC Application Focus	Apps- KnowledgeC Apps Focus • Bundle ID • Generated UUID • End Date • Category • Created • Data Provenance • Usage In Seconds • Start Time • Entry Creation • Path • Provenance
KnowledgeC Application Usage	Apps- KnowledgeC Apps Usage • Bundle ID • Start Time • Entry Creation • Path • Provenance • Usage in Seconds • End Date • Category • Created • Data Provenance

Device Interaction

Description	This section will contain information relating to device interactions which can be discovered and recovered using the Forensic Tool Kit. This includes information such as USB and Bluetooth connections, as well as peripheral device information. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, last connect, device name, and product vendor may be found under the Peripheral Device Information attribute.
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Attribute	Associated Columns
USB Devices	USB Devices - User Name - Last Removal - Vendor ID - Product ID Instance ID - Revision ID - First Connect Local - Volume Labels - Friendly Name - Cateogry - Provenance - Last Connect - Vendor - Product - Revision - Last User Mount - Mount Location - First Connect - Path - Created - Data Provenance
	Device-USB Connection History - Serial# - Vendor ID, Product ID - Path - Created - Data Provenance - Connection Start - Dev Release - Category - Provenance
USB Connections History	

Attribute	Associated Columns
AirDrop Discoverability	Device-Airdrop Discoverability (Mac) - Mode Change Time - Transact Log - Category - Provenance - Mode - Path - Created - Data Provenance
AirDrop Background Activity	Device-AirDrop Background Activity (Mac) - Date/Time - Category - Created - Data Provenance - Transact Log - Path - Provenance
Bluetooth Devices	Device-Bluetooth Devices (Mac) - Bluetooth Addr - Connect Type - Path - Created - Data Provenance - Dev Name - Last Seen - Category - Provenance
Peripheral Device Information	Device-Peripheral Information (Mac) - Dev Name - Built-in? - Vendor ID - Registry Entry ID - Category - Provenance - Last Connect - Connected - Product ID - Path - Created - Data Provenance

Browsers

Description	This section will contain information relating to browser information items which can be discovered and recovered using the Forensic Tool Kit. This includes application information such as downloads, searches, and history; also found here is information concerning categories of URLs. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, if looking at the Phishing URLs attribute, present nodes would include URL Title and Last Visit Time.
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Attribute	Associated Columns
	Browser-iCloud Downloads
iCloud Downloads	- File Size - Category - Created - Data Provenance - Download Request Time - Path - Provenance
	Browser-iCloud Local Files
iCloud Local Files	- Pkg Name - Created - Category - Data Provenance - File Size - Modified - Path Provenance
	Browser-iCloud Uploads
iCloud Uploads	- File Size - Category - Created - Data Provenance - Upload Request Time - Path - Provenance
	Browser-KnowledgeC Safari History
KnowledgeC Safari History	- Vol UUID - Content Desc - Title - Bundle ID - Source - Generated UUID

Attribute	Associated Columns
	- Start Time - Entry Creation - Path - Provenance - End Date - Category - Created - Data Provenance
Keyword Searches	Browser-Keyword Searches - User Name - Search Term - Web Page Title - Search Engine - Category - Provenance - Date/Time - URL - Browser - Path - Created - Data Provenance
Credentials	Browser-Credentials - Name - User Name - Browser - Path - Created - Data Provenance - Last Visit - URL - Password - Category - Provenance
Downloads	Browser-Downloads - User Name - URL - File Size - Path - Created - Data Provenance - File Path - Download Time - Browser - Category - Provenance
Cookies	Browser-Cookies - User Name - Value - Name - Last access

Attribute	Associated Columns	
	• Creation Time • Domain • Expires • Category • Provenance	• Browser • File Path • Path • Created • Data Provenance
Social Media URLs	Browser-URLs • Name • URL • Category • Description • Metadata • Category • Provenance	
Darknet URLs	Browser-URLs • Name • URL • Category • Description • Metadata • Category • Provenance	
Cryptocurrency URLs	Browser-URLs • Name • URL • Category • Description • Metadata • Category • Provenance	

Attribute	Associated Columns	
Tax Site URLs	Browser-URLs	
	• Name	• Last Visit
	• URL	• URL Title
	• Category	• Sub Category
	• Description	• Activity
	• Metadata	• Path
	• Category	• Created
	• Provenance	• Data Provenance
Google Searches	Browser-URLs	
	• Name	• Last Visit
	• URL	• URL Title
	• Category	• Sub Category
	• Description	• Activity
	• Metadata	• Path
	• Category	• Created
	• Provenance	• Data Provenance
Phishing URLs	Browser-URLs	
	• Name	• Last Visit
	• URL	• URL Title
	• Category	• Sub Category
	• Description	• Activity
	• Metadata	• Path
	• Category	• Created
	• Provenance	• Data Provenance
Pornography URLs	Browser-URLs	
	• Name	• Last Visit
	• URL	• URL Title
	• Category	• Sub Category

Attribute	Associated Columns
	• Description • Metadata • Category • Provenance • Activity • Path • Created • Data Provenance
Unknown URLs	Unknown URLs • Name • URL • Category • Description • Metadata • Category • Provenance • Last Visit • URL Title • Sub Category • Activity • Path • Created • Data Provenance

Networks

Description	This section will contain information relating to networks which can be discovered and recovered using the Forensic Tool Kit. This includes application information such as network profiles and interfaces. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, MAC, BSD Name, Network Type may be found under the Network Interfaces – MAC attribute.
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Attribute	Associated Columns
	Network-Utilities (Mac)
Network Utilities - MAC	- Utility Tool - Modified - Category - Provenance - Search Qry - Path - Created - Data Provenance
	Network-Profiles (Mac)
Network Profiles - MAC	- SSID - Last Connect - Path - Created - Data Provenance - MAC - Status - Category - Provenance
	Network-Interfaces (Mac)
Network Interfaces - MAC	- Network Name - BSD Name - Active - Srv Path - Category - Provenance - MAC - Network Type - USB Product Name - Path - Created - Data Provenance

Attribute	Associated Columns	
Daily Logs - Network Interface Status - MAC	Network Interface Status	
	• MAC • Errors In • MTU • Network Name • Packets Out • Path • Provenance	• Collision • Packets In • Name • Errors Out • Category • Created • Data Provenance

Recent Files/Folders

Description	This section will contain information relating to recently used files and folders which can be discovered and recovered using the Forensic Tool Kit.
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Attribute	Associated Columns
Bash/ZSH Sessions	Recent-Bash/ZSH Session (Mac)
	- User Name - Start Time - Path - Created - Data Provenance
	- Sess ID - End Date - Category - Provenance
Finder MRU - MAC	Recent-Finder MRU (Mac)
	- Creator Name - MRU Type - Category - Provenance
	- Fldr Path - Path - Created - Data Provenance
Recently Used Items	Recent-Recently Used Items (Mac)
	- File Name - MRU Type - Creation Time - Vol UUID - Volume Created - Vol Name - Provenance
	- Fldr Path - Access Order - Generated UUID - Last Access - Creator - Vol Size - Data Provenance
Spotlight Shortcuts	Recent-Spotlight Shortcut (Mac)
	- Search Qry - URL - Path - Created - Data Provenance
	- Display Name - Last Access - Category - Provenance

Operating System Information

Description	This section will contain information relating to operating system information which can be discovered and recovered using the Forensic Tool Kit. This includes features such as owner and account information. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, username, password hashes, and home directories may be found under the User Accounts attribute.
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Attribute	Associated Columns
Owner Information - MAC	Owner Information - MAC
	- Product Name - Product Version - Prod Major Vers - Time Zone - Installed - IOS Major Vers - Category - Created - Data Provenance - Prod C - Prod User Visible Vers - Computer Name - Curr Build # - Country Code - IOS Supported Vers - Path - Provenance
Apple Accounts	OS-Apple Accounts (Mac)
	- User Name - Acct ID - Parent Acct ID - Acct Type - Owning BundleID - Path - Created - Data Provenance - Local Account - Account Added - Acct Desc - Acct Cred Type - Last Credential Expiry - Category - Provenance

Attribute	Associated Columns	
User Accounts - MAC	OS-User Accounts (Mac) - User Name - Usr ID - Profile Creation - Pwd Hash - Pwd Hint - Last Bad Pwd Login - Profile Img - Path - Created - Data Provenance	
	- Full Name - Vol UUID - Home Dir - Pwd Hash Algorithm - Login Failure Cnt - Last Pwd Change - Profile Img Path - Category - Provenance	
Deleted Accounts	OS-Deleted Accounts (Mac) - User Name - Vol UUID - Delete Date - Category - Provenance	
Homebrew Cask Application	- Name - Item Number - System Info Version - System Info Description - System Info URL - System Info Provenance	
Homebrew Information	Normal - Name - Item Number - Path - Physical Size	
	- Label - Extension - File Type - Logical Size	

Attribute	Associated Columns	
	- MD5 Hash - SHA256 Hash - Accessed Date	- SHA1 Hash - Created Date - Modified Date
	Homebrew Build Dependency	
	- Name - Item Number - Full Name - Data Provenance	- Label - Homebrew Entry Name - System Info Provenance
	Homebrew Entry	
	- Name - Item Number - System Info Version - Compiler - Built On OS - Built On CPU Family - Built On OS CLT - Homepage - SHA 256 - Poured From Bottle - Installed As Dependency - System Info InstallTime - Changed Files - URL Mirrors - Data Provenance	- Label - System Info Application Name - Home Brew Version - Architecture - Built On OS Version - Built On OSX Code - System Info Description - System Info URL - Built As Bottle - Loaded From API - Installed On Request - Source Modification Time - Aliases - System Info Provenance

Attribute	Associated Columns
	Homebrew Resources • Name • Item Number • Full Name • SHA 256 • Data Provenance • Label • Homebrew Entry Name • System Info URL • System Info Provenance
	Homebrew Runtime Dependency • Name • Item Number • System Info Application • Declared Directly • Data Provenance • Label • Homebrew Entry Name • System Info Version • System Info Provenance

Other KnowledgeC, PowerLog & Logs Data

Description	This section will contain information relating to KnowledgeC, PowerLog & Logs Data which can be discovered and recovered using the Forensic Tool Kit. This includes device information such as battery and activity level. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, disk capacity, file system, and available disk may be found under the Daily Logs - Disk Status attribute.
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Attribute	Associated Columns
	PowerLog Timezone Information
PowerLog Timezone Information	- Timestamp - GMT Offset - Category - Provenance - Country Code - Path - Created - Data Provenance
	PowerLog Process Usage
PowerLog Process Data Usage	- Proc Name - Start Time - WiFi Bytes Sent - Mobile Bytes Sent - Path - Created - Data Provenance - Owning Bundle ID - End Date - WiFi Bytes Rcvd - Mobile Bytes Rcvd - Category - Provenance
	PowerLog Battery Level
PowerLog Battery Level	- Timestamp - Raw Batt Level - Full Charge - Path - Provenance - Batt Level - Charging - Category - Created - Data Provenance

Attribute	Associated Columns
KnowledgeC Activity Level	KnowledgeC Activity Level - Activity - End Date - Category - Created - Data Provenance - Start Time - Recorded - Path - Provenance
KnowledgeC Notification Usage	KnowledgeC Notification Usage - BundleID - Proc Name - End Date - Category - Created - Data Provenance - Type - Start Time - Recorded - Path - Provenance
KnowledgeC Screen Backlight States	KnowledgeC Screen Backlight States - Status - End Date - Category - Created - Data Provenance - Start Time - Recorded - Path - Provenance
Daily Logs - Disk Status	Daily Logs - Disk Status - Disk Capacity - Disk Space Available - Mount Point - Path - Provenance - File System - Disk Space Used - Category - Created - Data Provenance

Attribute	Associated Columns
Daily Logs - Local System Status	Daily Logs - Local System Status • Usr • Load Avg Last 5 min • Uptime • Path • Provenance • Load Avg Last min • Load Avg Last 15min • Category • Created • Data Provenance

Apple KeyChain Generic Password

Description	This section will contain information relating to Apple KeyChain items which can be discovered and recovered using the Forensic Tool Kit.
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Attribute	Associated Columns
Apple KeyChain Generic Password	Apple KeyChain Generic Password • Usr Acct • PW/Value • Service • Modified Category • Created • Data Provenance • Display Name • Creation Time • Record# • Path • Provenance

5 Mobile Forensics - Android

Application Usage

Description	This section will contain information relating to mobile application items which can be discovered and recovered using the Forensic Tool Kit. These include application information such as install states and application permissions; also found here is information concerning Google Play Store. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, the application name, file path, and install dates may be found under the Installed Applications attribute.
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Attribute	Associated Columns	
Android Device Information	• Device ID • Brand • Product • Android Version • Locale ID • IMEI List • Location Services? • MAC	• Model • Product Device • Manufacturer • Timezone • Advertising Key • Factory Rest Time • Serial#
Android Usage History	• Log Type • Class • Activity Event Type • Data Provenance	• Pkg Name • Timestamp • Provenance
Android User Dictionary	• Word • Locale • Path • Description	• User Dict. Freq. • Category • Created

Attribute	Associated Columns	
Application Activity - Android	- Start Time - Bundle ID - Activity Type - User Activity String - Invocation Phrase - App ID Activity UUID - Dat of Week - Entry Creation - Generated UUID - Assoc URL - Provenance	- End Time - Group - Content Desc - Content URL - Unique ID Source - Title - GMT Offset - App Expiry - Window Title - Json Payload - Data Provenance
Application Permissions - Android	- Pkg Name - App Install Time - Associated Account - App Version - Install Path - Is From Google Playstore? - Denied Permissions List	- Is Auto Update Enable? - Last Updated - Title - Data Storage Path - Install Source - Granted Permissions List - Provenance
Application Power Usage	- Interval Start Time - Bundle ID - Bkg Time (s) - Bkg Location Usage (s) - Data Provenance	- Interval (m) - Screentime (s) - Bkd Aud Played (s) - Provenance
Google Play Application Details	- Search Term - Provenance	- Timestamp - Data provenance
Google Play Installed Applications	- Associated Account - Purchased	- Application Name - Data Delivery

Attribute	Associated Columns	
	• First Download • App Install Time • Bytes Total • Title • Is Auto Update Enabled? • Provenance	• Last Update • Install Reason • Client Event ID • Last Notified Ver • Desired Ver
Google Play Searches	• Search Term • Provenance	• Timestamp
Installed Applications	• App • App ID • Permissions • App GUID • App Description • App Vendor • Path • Created	• App Version • App Install Time • App Update Timestamp • App Category • App Deleted Date • Copyright • Category • Description

Communication

Description	This section will contain information relating to general communication items which can be discovered and recovered using the Forensic Tool Kit. These include generic mobile features such as contact lists, SMS/MMS messages, and SIM card information; also found here is information concerning communications and messaging applications. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, names, addresses, and phone numbers may be found under the Contacts attribute.
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Attribute	Associated Columns
AIM Buddies	- Phone Number - First Name - Mobile Number - Work Number - Address - Address2 - Zip Src - Path - Description - Last Name - Number Type - Home Number - Email - Address1 - State - Category - Created
Android Call Logs	- ID, Phone Number - Via Number - Duration (s) - Subscription ID - Number Type - Country ISO - Contact ID - Post Dial Digits - Timestamp - Type - Contact Name - Number Label - Geocoded Location - Provenance

Attribute	Associated Columns
Android Contacts	• Phone Number • First Name • Mobile Number • Work Number • Address • Address2 • Zip Src • Path • Description • Last Name • Number Type • Home Number • Email • Address1 • State • Category • Created
Android Kik Messenger Attachments	• File Name • File Size • File Type • External URL
Android Kik Messenger Contacts	• Phone Number • First Name • Mobile Number • Work Number • Address • Address2 • Zip Src • Path • Description • Last Name • Number Type • Home Number • Email • Address1 • State • Category • Created
Android Kik Messenger Messages	• Sender • Time Sent • Category • Msg App • Message • Created • Recipient(s) • Time Received • Read • # Attachment • Path • Description

Attribute	Associated Columns	
Android Messages	• Sender • Time Sent • Category • Msg App • Message • Created	• Recipient(s) • Time Received • Read • # Attachment • Path • Description
Android MMS	• Subject • Sender • Sent • Attachments • MMS To • MMS BCC • Category • Description	• Message • Recipient(s) • Received • Src • MMS CC • Path • Created
Android Sim Card Information	• ICCID • IMSI • Display Name • MCC • MNC • SIM Change Time • Data Provenance	• Service Provider • SIM ID • Phone No • Country ISO • Provenance
Android SMS	• Description • SMS From Number • Message • Recipient(s) • Sent • Time Sent	• SMS Type • SMS To Number • Sender • Incoming • Read • Time Received

Attribute	Associated Columns	
	- SMS Time Zone of Sender - SMS Hdr. - Path - Created	- Src - SMSC Time - Category
Android SMS/MMS	- ID - Address - Timestamp - Read - Type - Direction - Provenance	- Thread ID - Subject - Sent - Seen SMS - Message - Creator
Android SMS/MMS (Content Provider)	- ID - Address - Timestamp - Read - Type - Direction - Provenance	- Thread ID - Subject - Time Sent - Seen SMS - Message - Creator - Data Provenance
Android SnapChat Photo Transfers	- File Name - File Size - Provenance	- File Type - External URL - Data Provenance
Android SnapChat Received Images	- File Name - File Size - Provenance	- File Type - External URL - Data Provenance
Android SnapChat Friends	- Phone Number - First Name	- Last Name - Number Type

Attribute	Associated Columns
	• Mobile Number • Work Number • Address • Address2 • Zip Src • Path • Description • Home Number • Email • Address1 • State • Category • Created
Discord Logged-in Account	• Profile Name • Service Type • Country • Category • Created • Profile • Password • Country Code • Path • Description
Discord Messages	• Sender • Time Sent • Category • Msg App • Message • Created • Recipient(s) • Time Received • Read • # Attachment • Path • Description
Facebook Messenger Calls	• Calling Name • Call Type • Call Duration • Src • Category • Created • Call Number • Call Time • Direction • Country • Path • Description
Facebook Messenger Groups	• Description • Convo Name • Chat Count • Category • Participants • Topic • Msg App • Path

Attribute	Associated Columns	
	Created	
Facebook Messenger Messages	- Sender - Time Sent - Category - Msg App - Message - Created	- Recipient(s) - Time Received - Read # Attachment - Path - Description
Facebook Messenger Users Contacted	- Phone Number - First Name - Mobile Number - Work Number - Address - Address2 - Zip Src - Path - Description	- Last Name - Number Type - Home Number - Email - Address1 - State - Category - Created
Google Duo Group Calls	- Description - Convo Name - Chat Count - Category - Created	- Participants - Topic - Msg App - Path
Google Duo Groups	- Description - Convo Name - Chat Count - Category - Created	- Participants - Topic - Msg App - Path

Attribute	Associated Columns	
Google Hangouts Cached Images	- File Name - File Size - Provenance	- File Type - External URL - Data Provenance
Google Hangouts Voice Calls	- Calling Name - Call Type - Call Duration - Src - Category - Created	- Call Number - Call Time - Direction - Country - Path - Description
Houseparty Messages	- Sender - Time Sent - Category - Msg App - Message - Created	- Recipient(s) - Time Received - Read # Attachment - Path - Description
Houseparty Users	- Profile Name - Service Type - Country - Category - Created	- Profile - Password - Country Code - Path - Description
imo Contacts	- Phone Number - First Name - Mobile Number - Work Number - Address - Address2 - Zip Src	- Last Name - Number Type - Home Number - Email - Address1 - State - Category

Attribute	Associated Columns
	• Path • Description • Created
imo Messages	• Sender • Time Sent • Category • Msg App • Message • Created • Recipient(s) • Time Received • Read • # Attachment • Path • Description
KakaoTalk Messages	• Sender • Time Sent • Category • Msg App • Message • Created • Recipient(s) • Time Received • Read • # Attachment • Path • Description
LINE Chats	• Description • Convo Name • Chat Count • Category • Created • Participants • Topic • Msg App • Path
LINE Contacts	• Phone Number • First Name • Mobile Number • Work Number • Address • Address2 • Zip Src • Path • Description • Last Name • Number Type • Home Number • Email • Address1 • State • Category • Created

Attribute	Associated Columns	
LINE Messages	• Sender • Time Sent • Category • Msg App • Message • Created	• Recipient(s) • Time Received • Read • # Attachment • Path • Description
LINE Pictures	• File Name • File Size • Provenance	• File Type • External URL • Data Provenance
Samsung Messages	• Sender • Time Sent • Category • Msg App • Message • Created	• Recipient(s) • Time Received • Read • # Attachment • Path • Description
Signal	• Created • Category • Path • L-Size	• Modified • Ext • MD5
Signal For Android	• Created • Category • Path • L-Size	• Modified • Ext • MD5
Signal Conversations - Android	• Description • Convo Name • Chat Count • Category	• Participants • Topic • Message App • Path

Attribute	Associated Columns	
	Created	
Signal Group Members	- Phone Number - First Name - Mobile Number - Work Number - Address - Address2 - Zip Src - Path - Description - Last Name - Number Type - Home Number - Email - Address1 - State - Category - Created	
Signal Groups	- Description - Convo Name - Chat Count - Category - Created - Participants - Topic - Msg App - Path	
Signal Local User	- Profile Name - Service Type - Country - Category - Created - Profile - Password - Country Code - Path - Description	
Signal Messages - Android	- Sender - Time Sent - Category - Msg App - Message - Created - Recipient(s) - Time Received - Read # Attachment - Path - Description	

Attribute	Associated Columns
Signal Users	• Profile Name • Service Type • Country • Category • Created • Profile • Password • Country Code • Path • Description
Skype Accounts	• Profile Name • Service Type • Country • Category • Created • Profile • Password • Country Code • Path • Description
Skype Calls	• Calling Name • Call Type • Call Duration • Src • Category • Created • Call Number • Call Time • Direction • Country • Path • Description
Skype Chat Messages	• Sender • Time Sent • Category • Msg App • Message • Created • Recipient(s) • Time Received • Read • # Attachment • Path • Description
Skype Chatsync Messages	• Sender • Time Sent • Category • Msg App • Message • Recipient(s) • Time Received • Read • # Attachment • Path

Attribute	Associated Columns	
	Created	Description
Skype Contacts	- Phone Number - First Name - Mobile Number - Work Number - Address - Address2 - Zip Src - Path - Description	- Last Name - Number Type - Home Number - Email - Address1 - State - Category - Created
Skype File Transfers	- File Name - File Size - Provenance	- File Type - External URL - Data Provenance
Skype Group Chat	- Description - Convo Name - Chat Count - Category - Created	- Participants - Topic - Message App - Path
Snapchat Chat Messages	- Sender - Time Sent - Category - Msg App - Message - Created	- Recipient(s) - Time Received - Read # Attachment - Path - Description
Snapchat Received Videos	- File Name - File Size - Provenance	- File Type - External URL - Data Provenance

Attribute	Associated Columns	
Telegram Chats - Android	• Description • Convo Name • Chat Count • Category • Created	• Participants • Topic • Message App • Path
Telegram Contacts - Android	• Phone Number • First Name • Mobile Number • Work Number • Address • Address2 • Zip Src • Path • Description	• Last Name • Number Type • Home Number • Email • Address1 • State • Category • Created
Telegram Messages - Android	• Sender • Time Sent • Category • Msg App • Message • Created	• Recipient(s) • Time Received • Read • # Attachment • Path • Description
Telegram Users - Android	• Profile Name • Service Type • Country • Category • Created	• Profile • Password • Country Code • Path • Description
Viber Messages	• Sender • Time Sent	• Recipient(s) • Time Received

Attribute	Associated Columns	
	• Category • Msg App • Message • Created	• Read • # Attachment • Path • Description
WeChat Friends	• Phone Number • First Name • Mobile Number • Work Number • Address • Address2 • Zip Src • Path • Description	• Last Name • Number Type • Home Number • Email • Address1 • State • Category • Created
WeChat Messages	• Sender • Time Sent • Category • Msg App • Message • Created	• Recipient(s) • Time Received • Read • # Attachment • Path • Description
WhatsApp	• Created • Category • Path • L-Size	• Modified • Ext • MD5
WhatsApp for Android	• Created • Category • Path • L-Size	• Modified • Ext • MD5

Attribute	Associated Columns	
Android WhatsApp Accounts Information	• Profile Name • Service Type • Country • Category • Created	• Profile • Password • Country Code • Path • Description
Android WhatsApp Chats	• Description • Convo Name • Chat Count • Category • Created	• Participants • Topic • Msg App • Path
Android WhatsApp Contacts	• Phone Number • First Name • Mobile Number • Work Number • Address • Address2 • Zip Src • Path • Description	• Last Name • Number Type • Home Number • Email • Address1 • State • Category • Created
Android WhatsApp Groups	• Description • Convo Name • Chat Count • Category	• Participants • Topic • Msg App • Path Created
Android WhatsApp Live Locations	• Location Name • Device Long. • Location Country • Location Address	• Device Lat. • City • Location Category • Category

Attribute	Associated Columns	
	• Path • Description	• Created
Android WhatsApp Messages	• Sender • Time Sent • Category • Msg App • Message • Created	• Recipient(s) • Time Received • Read • # Attachment • Path • Description
Android WhatsApp Profile Pictures	• File Name • File Size • Provenance	• File Type • External URL • Data Provenance
Android WhatsApp User Profiles	• Profile Name • Service Type • Country • Category • Created	• Profile • Password • Country Code • Path • Description

Connected Devices

Description	This section will contain information relating to items connectable to the mobile device, which can be discovered and recovered using the Forensic Tool Kit. These include health tracking devices, recording steps and heart rates; also found here is information concerning general Bluetooth and SIM card activity. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found.
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Attribute	Associated Columns
Fitbit Floors	- DataSource Model - End Time - Start Time - Number of Floors
Fitbit Heart Rate	- DataSource Model - End Time - Provenance - Start Time - Measurement - Data Provenance
Fitbit Steps	- Start Time - Number of Steps - End Time
Fitbit Profiles	- Profile Name - Service Type - Country - Category - Created - Profile - Password - Country Code - Path - Description
Samsung Health Steps (Device)	- Start Time - Number of Steps - End Time
Samsung Health Steps (Wearable)	- Start Time - Number of Steps - End Time
Samsung Health User Profiles	- Profile Name - Service Type - Country - Category - Created - Profile - Password - Country Code - Path - Description

Attribute	Associated Columns	
Bluetooth Devices	- Hw Address - Type - Last Name Updt	- Name - Last Svc Updt - Provenance
SIM Card Activity	- ICCID - Last Update - Home Budget	- Number - Slot ID - Roaming Budget
SIM Card ICCID	- ICCID - Slot ID - Roaming Budget - Data Provenance	- Number Last Update - Home Budget - Provenance
SIM Card IMSI	- ICCID - IMSI - Display Name - MCC MNC - SIM Change Time - Data Provenance	- Service Provider - SIM ID - Phone No - Country ISO - Provenance
SIM Card Phone Numbers	- ICCID - Slot ID - Roaming Budget - Data Provenance	- Number Last Update - Home Budget - Provenance
SIM Card Service Providers	- ICCID - IMSI - Display Name - MCC MNC - SIM Change Time - Data Provenance	- Service Provider - SIM ID - Phone No - Country ISO - Provenance
SIM Card SMS Messages	- Description - SMS From Number	- SMS Type - SMS To Number

Attribute	Associated Columns
	• Message • Recipient(s) • Sent • Time Sent • SMS Time Zone of Sender • SMS Hdr. • SMSC Time • Category
	• Sender • Incoming • Read • Time Received • Src • SMS Country • Path • Created

Email and Calendar

Description	This section will contain information relating to general email and calendar items which can be discovered and recovered using the Forensic Tool Kit. These include emails sent and received from providers such as Gmail and Yahoo; also found here is calendar information, again from providers such as Google, and those included as built-in applications featured on an Android mobile device. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, names, email addresses, and message bodies may be found under the Emails attribute.
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Attribute	Associated Columns	
Android Emails	• Thread ID • Message Row ID • To • BCC • Snippet • Timestamp	• ID • From • CC • Subject • Message Body
Android Gmail Conversations	• Description • Convo Name • Chat Count • Category • Created	• Participants • Topic • Message App • Path
Android Yahoo Mail Attachments	• File Name • File Size	• File Type • External URL
Android Yahoo Mail Emails	• Thread ID • Message Row ID • To • BCC • Snippet	• ID • From • CC • Subject • Message Body

Attribute	Associated Columns	
	• Timestamp	
Android Yahoo Mail User Accounts	• Profile Name • Service Type • Country • Category • Created	• Profile • Password • Country Code • Path • Description
Calendar Events	• ID • Cal Description • Latitude • Deleted • Is All Day? • End Time • Map Bytes • Organiser	• Title • Location • Longitude • Alarm? • Start Time • Last Time • Self Attendee? • Provenance
Calendar Events (UFED Agent)	• ID • Cal Description • Latitude • Deleted • Is All Day? • End Time • Map Bytes • Organizer • Data Provenance	• Title • Location • Longitude • Alarm? • Start Time • Last Time • Self Attendee? • Provenance
Gmail Emails	• Thread ID • Message Row ID • To • BCC	• ID • From • CC • Subject

Attribute	Associated Columns	
	• Snippet • Timestamp	• Message Body
Google Calendar Calendars	• Acct ID • Calendar Name • Has Owner Access? • Internal Data Location	• Calendar ID • Calendar Details • To Be Removed? • Provenance
Google Calendar Events	• Event ID • Description • End Time • URL	• Event Name • Start Time • Creation Time • Provenance
ProtonMail Contacts	• Phone Number • First Name • Mobile Number • Work Number • Address • Address1 • State • Category	• Last Name • Number Type • Home Number • Email • Work Address • Address2 • Zip Src • Path
ProtonMail Emails	• Thread ID • Message Row ID • To • BCC • Snippet • Timestamp	• ID • From • CC • Subject • Message Body
Samsung Email Logs	• Thread ID • Message Row ID • To	• ID • From • CC

Attribute	Associated Columns
	• BCC • Snippet • Timestamp • Data Provenance • Subject • Message Body • Provenance

Encryption and Credentials

Description	This section will contain information relating to Android KeyStore items which can be discovered and recovered using the Forensic Tool Kit.
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Attribute	Associated Columns	
Android KeyStore	• Result Type • Description • Decrypted Data	• Application Name • Encrypted Data • Provenance
Android KeyStore - GrayKey	• Result Type • Description • Decrypted Data • Data Provenance	• Application Name • Encrypted Data • Provenance

Location and Travel

Description	This section will contain information relating to location and travel items which can be discovered and recovered using the Forensic Tool Kit. These include mobile map applications such as Google Maps, as well as mobility service providers such as Uber. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, locations, directions, and addresses may be found under the Locations attribute.
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Attribute	Associated Columns	
Android Google Maps	• Ext • Category • L-Size • SHA1 • Created • Modified	• Path • P-Size • MD5 • SHA256 • Accessed
Android Wi-Fi Profiles	• SSID • MAC • Last Connect • Updated Time • Creator UID • Ephemeral? • Valid Internet Access? • Has Ever Connected? • Data Provenance	• WiFi Security • Connection Time List • Creation Time • Connection Priority • Network ID • SelfAdded? • Shared? • Provenance
Google Maps Directions	• Timestamp • Estimated Travel • Data Provenance	• Journey Steps Information • Provenance
Last Known Locations	• Location Name	• Device Lat.

Attribute	Associated Columns
	• Device Long. • Location Country • Location Address • Path • Description • City • Location Category • Category • Created
Uber Accounts	• Profile Name • Service Type • Country • Category • Created • Profile • Password • Country Code • Path • Description
Uber Cached Locations	• Location Name • Device Long. • Location Country • Location Address • Path • Description • Device Lat. • City • Location Category • Category • Created
Uber Payments	• Creation Time • Timestamp • Amount • Merchant Industry • Merchant Category • Location Time • Longitude • Horiz Accuracy • Provenance • Modified • Organization Name • Currency Code • Merchant Name • Serial# • Latitude • Elevation • Vert Accuracy • Data Provenance
Uber Profiles	• Profile Name • Service Type • Country • Profile • Password • Country Code

Attribute	Associated Columns
	• Category • Created • Path • Description
Waze Places	• Location Name • Device Long. • Location Country • Location Address • Path • Description • Device Lat. • City • Location Category • Category • Created

Media

Description	This section will contain information relating to media items which can be discovered and recovered using the Forensic Tool Kit. These include generic mobile features such as pictures and videos; also found here is information concerning photo sharing and storage services such as Google Photos. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, capture dates, makes and models of the device used, may be found under the Pictures attribute.
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Attribute	Associated Columns	
AMR Files		
Audio	- Created - L-Size - Path	- Modified - Category
Carved Video	- Created - L-Size - Capture Date - Model - Longitude	- Modified - Category - Make - Latitude - Path
Google Photos Albums	- ID - UTC Time - Capture Time - Type - Is VR? - Hidden - Partial Backup Downloaded - OEM Special Type - In Camera Folder? - Overlay Type - Date Header UTC Time	- Ded Up Key - GMT Offset - Month Random Time - Deleted - Has Local? - Composition Type - Is Archive? - Is Favourite? - In Primary Stage? - Min Upload UTC Time - Trash Time

Attribute	Associated Columns	
Google Photos Media	- Type - ID - UTC Time - Capture Time - Type - Is VR? - Hidden - Partial Backup Downloaded - OEM Special Type - In Camera Folder? - Overlay Type - Date Header UTC Time - Type	- Provenance - Ded Up Key - GMT Offset - Month Random Time - Deleted - Has Local? - Composition Type - Is Archive? - Is Favourite? - In Primary Stage? - Min Upload UTC Time - Trash Time - Provenance
Motion Photos	- Capture Date - Model - Longitude - Duplicate File - Category	- Make - Latitude - L-Size - Path - Created
Pictures	- Capture Date - Model - Longitude - Duplicate File - Category	- Make - Latitude - L-Size - Path - Created
Videos	- Created - L-Size - Capture Date - Model - Longitude	- Modified - Category - Make - Latitude - Path

Mobile Application Attachment

This is applicable for Microsoft Teams, Slack and LinkedIn.

Attribute	Associated Columns
Attachment	- Label - System Info File Name - System Info File Size - Item Source - System Info Provenance - Name - Item Number - Summary MD File Type - System Info Target Creation Time - System Info Source - Data Provenance

Mobile Contacts

This is applicable for Microsoft Teams, Skype, Slack, LinkedIn, and TikTok.

Attribute	Associated Columns
Contacts	- Label - Phone Number - Last Name - Mobile Number - Work Number - Address - Address1 - State - Item Source - Path - Item Description - Item Number - First Name - Number Type - Home Number - Email - Work Address - Address2 - Zip - File Type - Created Date - Name

Mobile Chat Conversations

This is applicable for Microsoft Teams, Skype, Slack, LinkedIn, and TikTok.

Attribute	Associated Columns	
Conversations	- Label - Item Description - Conversation Name - Chat Message Count - File Type - Created Date	- Item Number - Chat Participants - Chat Topic - Message Application - Path - Name

Mobile Calls

This is applicable for Microsoft Teams and Skype.

Attribute	Associated Columns	
Calls	- Label - Calling Name - Message Sender - Call Type - Call Duration - Call Result Status - Call Country Code - Path - Item Description	- Item Number - Call Number - Message Recipient(s) - Call Time - Direction - Item Source - File Type - Created Date - Name

Mobile Messages

This is applicable for Microsoft Teams, Skype, Slack, LinkedIn, and TikTok.

Attribute	Associated Columns	
Messages	• Label • Message Sender • Message Sent Time • File Type • Message Application • Message Body • Created Date • Name	• Item Number • Message Recipient(s) • Message Received Time • Message Is Read • Message Attachment Count • Path • Item Description

Operating System

Description	This section will contain information relating to operating system information which can be discovered and recovered using the Forensic Tool Kit. These include features such as android accounts information, downloads, and WiFi logs. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, SSID and connection times, may be found under the Wi-Fi Logs attribute.
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Attribute	Associated Columns	
Accounts Information	Type	Password
FCM (Firebase Cloud Messaging)	- Name - Item Number - Package Name - System Info Provenance	- Label - Timestamp Date/Time - Record ID - Data Provenance
Android Downloads	- File Path - HTTP Status - Notification Package - Creator UID - Description - Is Visible Download UI? - Deleted - Request Header URI	- MIME - Last Update - Bytes Total - Title - Is Public API? - Media Provider URI - Error Msg - Remote URI
Google Accounts	- Profile Name - Service Type - Country - Category - Created	- Profile - Password - Country Code - Path - Description

Attribute	Associated Columns
Wi-Fi Logs - Android	• SSID • MAC • Last Connect • Updated Time • Creator UID • Ephemeral? • Valid Internet Access? • Has Ever Connected? • Data Provenance • WiFi Security • Connection Time List • Creation Time • Connection Priority • Network ID • SelfAdded? • Shared? • Provenance

Social Networking

Description	This section will contain information relating to general social media items which can be discovered and recovered using the Forensic Tool Kit. These include mobile social media features for a variety of application, such as application users, messages, and contacts. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, profile names and passwords may be found under the Profiles/Users attributes.
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Attribute	Associated Columns	
Android Instagram Users	- Profile Name - Service Type - Country - Category - Created	- Profile - Password - Country Code - Path - Description
Android Tinder Accounts	- Profile Name - Service Type - Country - Category - Created	- Profile - Password - Country Code - Path - Description
Android Tinder Messages	- Sender - Sent - Category - Msg App - Message - Created	- Recipient - Received - Read # Attachments - Path - Description
Android Tinder Photos	- File Name - File Size - Provenance	- File Type - External URL - Data Provenance

Attribute	Associated Columns
Facebook Android Messages	• Sender • Sent • Category • Msg App • Message • Created • Recipient • Received • Read • # Attachments • Path • Description
Facebook Android Pictures	• File Name • File Size • Provenance • File Type • External URL • Data Provenance
Facebook Contacts	• Phone Number • First Name • Mobile Number • Work Number • Address • Address1 • State • Category • Created • Last Name • Number Type • Home Number • Email • Work Address • Address2 • Zip Src • Path • Description
Facebook User/Friends	• Phone Number • First Name • Mobile Number • Work Number • Address • Address2 • Zip Src • Path • Description • Last Name • Number Type • Home Number • Email • Address1 • State • Category • Created

Attribute	Associated Columns	
Instagram Direct Messages	• Sender • Time Sent • Category • Msg App • Message • Created	• Recipient(s) • Time Received • Read • # Attachments • Path • Description
Instagram Group Members	• Description • Convo Name • Chat Count • Category • Created	• Participants • Topic • Msg App • Path
Instagram Media	• File Name • File Size • Provenance	• File Type • External URL • Data Provenance
Instagram Profiles	• Profile Name • Service Type • Country • Category • Created	• Profile • Password • Country Code • Path • Description
Musical.ly Local Users	• Profile Name • Service Type • Country • Category • Created	• Profile • Password • Country Code • Path • Description
Musical.ly Messages	• Sender • Sent • Category	• Recipient • Received • Read

Attribute	Associated Columns	
	• Msg App • Message • Created	• # Attachments • Path • Description
Musical.ly Posts	• File Name • File Size • Provenance	• File Type • External URL • Data Provenance
Musical.ly Users	• Profile Name • Service Type • Country • Category • Created	• Profile • Password • Country Code • Path • Description
Reddit Accounts	• Profile Name • Service Type • Country • Category • Created	• Profile • Password • Country Code • Path • Description
Tumblr Chat Messages	• Sender • Sent • Category • Msg App • Message • Created	• Recipient • Received • Read • # Attachments • Path • Description
Twitter Direct Messages	• Sender • Sent • Category • Msg App • Message	• Recipient • Received • Read • # Attachments • Path

Attribute	Associated Columns
	• Created • Description
Twitter Users	• Profile Name • Profile • Service Type • Password • Country • Country Code • Category • Path • Created • Description

Twitter

Attribute	Associated Columns
Users	• Label • User ID • Image URL • DMS Muted • NFT Avatar • Super Followers Count • Verified Phone • System Info Description • URL Entity • Structured Elements • Fast Followers • Statuses • Media Count • Profile Update Time • Advertiser Type • Tip JAR Settings • Translator Type • User Label Data Elements • Friendship Time • Data Provenance • Item Number • System Info User Name • User Flags • Active Creator • Blue Verified • Twitter Articles • Header URL • Web URL • Location • Followers • Friends • Favourites • Profile Creation Date/Time • Pinned Tweet Time • Professional Elements • Hash • Profile Interstitial Type • Friendship • System Info Provenance • Name

Attribute	Associated Columns
Tweets	• Label • User ID • Status ID • Created Time • In status ID • Favourites • Retweet Count • View Count • Latitude • Place Data Element • Unified Card Element • Conversation Control Element • Reaction Meta Element • Eligibility Element • System Info Language • Reply Count • Self Thread ID • Composer Source • Limited Action • Community ID • Trusted Friend Screen Name • Unmentioned Info Elements • Rent Content Elements • System Info Provenance • Name • Item Number • System Info User Name • Content Element • In Ruser ID • In screen Name • Favourites • Quote Count • Flags • Longitude • Card Element • Perma Link Element • Voice Information Element • Community Element • Control Element • Quoted Tweet ID • Conversation ID • Reported • Tweet Source • Bird Watch Nodes • Down Voted • Quick Promote Elements • Content Elements • Retweeted • Data Provenance

Attribute	Associated Columns	
Live Events	• Label • Event ID • Date Time (String) • Short Title • Refresh Rate • Social Context Elements • Carousel Elements • Hash Tag • Description Entities • Data Provenance	• Item Number • Sensitive • System Info Description • Updated Time • System Info Category • Live Sports Score Elements • Timeline Elements • System Info Title • System Info Provenance • Name
Interest Topic	• Label • Topic ID • Following • Not Interested • System Info Provenance • Name	• Item Number • Topic ID Hash • System Info Description • Icon URL • Data Provenance
Information / User Information	• Name • Item Number • Modified Date • Extension • MD5 Hash	• Label • Created Date • File Type • Path • Logical Size

Uber

Attribute	Associated Columns	
Account	• Label	• Item Number
	• Device UUID	• Last Modified Time
	• Origin Time	• First Name
	• Last Name	• Email
	• Owner Picture URL	• Admin
	• Confirmed Mobile	• Confirmed Mobile Status
	• SMS Notification	• Country ISO
	• Mobile Digits	• No Password
	• Payment Profile UUID	• Payment Profile Google Wallet
	• Referral Code	• System Info Referring URL
	• Role	• Profile Type
	• Rating	• Confirmed Email
	• Riders Trip Count	• Latitude Value
	• Longitude Value	• System Info Provenance
	• Data Provenance	• Name
Cached Shortcuts	• Label	• Item Number
	• Timestamp Date/Time	• Device UUID
	• Full Address	• Latitude Value
	• Longitude Value	• Identifier
	• Locale	• System Info Provider
	• System Info Title	• Accelerator Type
	• Icon URL	• Tag Key
	• City ID	• Confidence(String)
	• Score(Double)	• Expiry Time
	• Sub Title	• Distance (m)
	• Analytics	• System Info Provenance

Attribute	Associated Columns	
Cached Location	• Data Provenance	• Name
	• Label	• Item Number
	• Timestamp Date/Time	• Uber ID
	• Reference ID	• Hash
	• Tag	• Title Segment
	• Subtitle Segment	• Relevance
	• Latitude Value	• Longitude Value
	• Total Trips	• Expiry Time
	• City ID	• Historical
	• System Info Provenance	• Data Provenance
	• Name	

Web Related

Description	This section will contain information relating to general web browser items which can be discovered and recovered using the Forensic Tool Kit. These include browser features such as browser history, bookmarks, and downloads; also found here is autofill information, such as credit cards and login details. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, searches and corresponding URLs and web page titles may be found under the Search Terms attributes.
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Attribute	Associated Columns	
Android Browser Bookmarks	- Created - Category - Path - L-Size	- Modified - Ext - MD5
Android Browser Search Terms	- User name - Search Term - Web Page Title - Search Engine - Category - Provenance	- Date/Time - URL - Browser - Path - Created
Android Browser Web History	- Visit Date - URL - Referring URL	- Visit Count - Title
Android Firefox Bookmarks	- Created - Category - Path - L-Size	- Modified - Ext - MD5
Android Firefox Web History	- Visit Date - URL	- Visit Count - Title

Attribute	Associated Columns	
	Referring URL	
Chrome Archived Keyword Search Terms	- User name - Search Term - Web Page Title - Search Engine - Category - Provenance	- Date/Time - URL - Browser - Path - Created
Chrome Autofill Profiles	- Object ID - Browser Origin - Value - Last Use Ft - Provenance	- Object Name - Name - Creation Ft - Count - Data Provenance
Chrome Autofill	- Object ID - Browser Origin - Value - Last Use Ft - Provenance	- Object Name - Name - Creation Ft - Count - Data Provenance
Chrome Bookmarks	- Created - Category - Path - L-Size	- Modified - Ext - MD5
Chrome Cookies	- User Name - Value - Creation Time - Domain - Expires - Category	- Name - Last Access - Browser - File Path - Path - Created

Attribute	Associated Columns	
Chrome Downloads	• User Name • URL • File Size • Path • Created	• File Path • Download Time • Browser • Category
Chrome Keyword Search Terms	• User name • Search Term • Web Page Title • Search Engine • Category • Provenance	• Date/Time • URL • Browser • Path • Created
Chrome Logins	• Login Row ID • Action URL • Username Value • Password Value	• Origin URL • Username Element • Password Element
Chrome Saved Credit Cards	• Card Type(s) • Expiration Year • Name on Card • Status • Data Provenance	• Expiration Month • Last 4 • Issuer • Provenance
Chrome Tab History	• ID • URL • Original Request URL • Transition Type	• Index Entry • Title • Referrer URL
Chrome Top SitesUs	• Browser • Title • Category • Provenance	• Builtin? • URL • Description • Data Provenance

Attribute	Associated Columns	
Chrome Web History	- Visit ID - Title - Provenance	- URL - Visit Count - Data Provenance
Chrome Web Visits	- Visit Date - URL - Referring URL	- Visit Count - Title
Edge Chromium Bookmarks	- Created - Category - Path - L-Size	- Modified - Ext - MD5
Edge Chromium FavIcons	- Browser - Last Update	Page URL
Edge Chromium Keyword Search Terms	- User Name - Search Term - Web Page Title - Search Engine - Category - Provenance	- Data/Time - URL - Browser - Path - Created - Data Provenance
Edge Chromium Tab History	- ID - URL - Original Request URL - Transition Type	- Index Entry - Title - Referrer URL
Edge Chromium Web History	- Visit ID - Title - Provenance	- URL - Visit Count - Data Provenance
Edge Chromium Web Visits	- Visit Date - URL - Referring URL	- Visit Count - Title

Attribute	Associated Columns	
Firefox Cookies	- Name - Last Access - Browser - File Path - Path - Created	- Value - Creation Time - Domain - Expires - Category
Firefox Web History	- Visit Date - URL - Referring URL	- Visit Count - Title
Firefox Web Visits	- Visit Date - URL - Referring URL	- Visit Count - Title
Opera Autofill	- Browser Origin - Value - Last Use - Provenance	- Name - Creation Time - Count - Data Provenance
Opera Bookmarks	- Created - Category - Path - L-Size	- Modified - Ext - MD5
Opera Cookies	- Name - Last Access - Browser - File Path - Path - Created	- Value - Creation Time - Domain - Expires - Category
Opera Downloads	User Name	File Path

Attribute	Associated Columns
	• URL • File Size • Path • Created • Download Time • Browser • Category
Opera FavIcons	• Browser • Last Update • Page URL
Opera Keyword Search Terms	• User name • Search Term • Web Page Title • Search Engine • Category • Provenance • Date/Time • URL • Browser • Path • Created
Opera Top Sites	• Browser • Title • Category • Provenance • Built-in? • URL • Description • Data Provenance
Opera Web History	• Visit Date • URL • Referring URL • Visit Count • Title
Opera Web Visits	• Visit Date • URL • Referring URL • Visit Count • Title
UC Browser Bookmarks	• Created • Category • Path • L-Size • Modified • Ext • MD5
UC Browser Cookies	• Name • Value

Attribute	Associated Columns	
	• Last Access • Browser • File Path • Path • Created	• Creation Time • Domain • Expires • Category
UC Browser Downloads	• User Name • URL • File Size • Path • Created	• File Path • Download Time • Browser • Category
UC Browser History	• Visit Date • URL • Referring URL	• Visit Count • Title

Mobile Privacy Dashboard

- Object Id
- Operation Type
- Bundle Name
- Data Provenance
- Object Name
- Activity Type
- Provenance
- Timestamp time

Mobile Intelligence Battery Usage

- Name
- Item Number
- Foreground Usage Time Millise
- ID
- User ID(Number)
- Package Name
- Battery Status
- Boot Time
- Total Power mAh
- Percentage Of Total Power
- Data Provenance
- Label
- Application Label
- Background usage Time Millise
- UID
- Battery Percentage
- Summary MD Timezone
- System Info Is Hidden
- Timestamp Date/Time
- Consumed Power mAh
- System Info Provenance

Recent Tasks

- Snapshot Image
- Object Name
- Real Activity
- Calling UID
- Calling Package
- Component
- Snapshot Image Path
- Data Provenance
- Object ID
- Task ID
- Affinity
- Effective UID
- User ID
- App Action
- Provenance
- Last Moved Time

Device Power

- Object Id
- Requester
- Timezone
- Data Provenance
- Utc Time
- Object Name
- Reason
- Provenance
- Local Time

Google Keep Application

This is applicable for **Google Keep Account**, **Google Keep Note** and **Google Keep Shared**.

- Name
- Item
- Modified Date
- Extension
- MD5
- Object Name
- Object ID
- Last Modified
- Physical Size
- Logical Size
- MD5 Hash
- SHA256 Value
- Label
- Created Date
- Category
- Path
- L-size
- Labels
- Created Date
- File Type
- Extension
- Path
- SHA1 Hash

Google Fit

- Name
- Pkg name
- Activity Name
- Type Activity
- Provenance
- Object Name
- Activity ID
- Activity String
- Summary Start Time
- Data Provenance
- Label
- Activity ID
- Activity String
- Start Time
- Data Provenance
- Labels
- Summary Description
- Activity Time
- End Time
- Status
- Item
- Description
- Activity Time
- End Time
- Object ID
- Pkg name
- Activity Name
- Type Activity
- Provenance
- Search Hits

Google Keyboard

- Name
- Training Target
- End Time
- Provenance
- Object Name
- Pkg name
- Session Number
- Data Provenance
- Label
- Pkg name
- Session Number
- Data Provenance
- Labels
- Summary Start time
- Locale
- Status
- Item
- Start time
- Locale
- Object ID
- Training Target
- End Date
- Provenance
- Search Hits

Garmin Health Fitness Application

Applicable for the below Sub-Nodes:

- Application Notification

- Cached Activity Information
 - Cached Activity

- Polyline Co-ordinates
 - Cached Activity summary
 - Sleep Event
 - Smart device
 - User daily Summary
 - User Profile
 - Weather Information

The following are the list of Columns:

- | | | |
|----------------|-----------------|-----------------|
| • Name | • Label | • Item |
| • Created Date | • Modified Date | • Category |
| • Extension | • Path | • MD5 |
| • L-size | • Object Name | • Labels |
| • Object ID | • Created Date | • Last Modified |
| • File Type | • Physical Size | • Extension |
| • Logical Size | • Path | • MD5 Hash |
| • SHA1 Hash | • SHA256 Value | |

Android turbo Health

Attribute	Associated Columns	
Mobile Turbo Application Launch Event	• Name • Item Number • Launchtime Value • Data Provenance	• Label • Package Name • System Info Provenance
Mobile turbo Battery Event	• Name • Item Number • Battery Level Value • Time zone • Data Provenance	• Label • Timestamp Date/Time • Charge Type • System Info Provenance

6 Mobile Forensics - iOS

Application Usage

Description	This section will contain information relating to mobile application items which can be discovered and recovered using the Forensic Tool Kit. This includes application information such as install states and application permissions; also found here are attributes concerning KnowledgeC application information. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, application name, file path, and install dates may be found under the Installed Applications attribute.
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Attribute	Associated Columns	
Application Install States	- Recorded - End Time - App Category - Type	- Start Time - Bundle ID - Application Name - Provenance
Application Permissions - MacOS, iOS	- Service - Perm Granted? - Provenance	- Service Type - Assoc User
Installed Applications	- Usr - Pkg Name - Install Date - Internal Vers - App Store Action - Bundle ID - CTR Moved - Dumped Loc	- Application Name - File Path - Display Vers - File Size - Publisher - CTR Made Live - CTR Path - Provenance
InteractionC Contacts	Creation Time	First Inbound Interact

Attribute	Associated Columns	
	• Last Inbound Interact • Last Outbound Interact • # Outbound Interacts	• First Outbound Interact • # Inbound Interacts • Address Display Name
InteractionC Interactions	• Ext • Category • L-Size • SHA1 • Created • Modified	• Path • P-Size • MD5 • SHA256 • Accessed
Biome App In Focus (iOS)	• Name • Item Number • Summary MD End Time • Bundle ID • Action GUID • System Info Provenance	• Label • Start Time • Write Time • Transition • System Info Activity • Data Provenance
Biome Application Install (iOS)	• Name • Item Number • Summary MD End Time • Bundle ID • System Info Activity • Data Provenance	• Label • Start Time • Write Time • Action GUID • System Info Provenance
Biome Application Intent (iOS)	• Name • Item Number • Summary MD End Time • Donated By Siri • Bundle ID	• Label • Start Time • System Info Duration • Identifier • Intent

Attribute	Associated Columns	
	- Intent Action - Group ID - System Info Provenance	- Direction(Number) - Intent Strings - Data Provenance
Biome Backlight (iOS)	- Name - Item Number - Backlight Active - Data Provenance	- Label - Timestamp Date/Time - System Info Provenance
Biome Battery Percentage (iOS)	- Name - Item Number - Summary MD End Time - Battery Percentage - System Info Activity - Data Provenance	- Label - Start Time - Write Time - Action GUID - System Info Provenance
Biome Device plugged (iOS)	- Name - Item Number - Summary MD End Time - Device Power Plugged - System Info Provenance	- Label - Start Time - Write Time - System Info Activity - Data Provenance
Biome Hardware Reliability (iOS)	- Name - Item Number - Hardware - Data Provenance	- Label - Timestamp Date/Time - System Info Provenance
Biome Notification (iOS)	- Name - Item Number - Bundle ID - Field2 - Field4 - Field6	- Label - Timestamp Date/Time - Field1 - Field3 - Field5 - System Info Provenance

Attribute	Associated Columns	
	Data Provenance	
Biome Text Input Session (iOS)	- Name - Item Number - Bundle ID - Data Provenance	- Label - Start Time - System Info Provenance
Biome User Activity (iOS)	- Name - Item Number - Activity Type - Field2 - System Info Provenance	- Label - Start Time - Field1 - Field3 - Data Provenance
Biome WiFi (iOS)	- Name - Item Number - Summary MD End Time - System Info Activity - Data Provenance	- Label - Start Time - Device Name - System Info Provenance
iTunes Metadata (iOS)	- Name - Item Number - Bundle Short Version String - Account Alt DSID - Account DS Person ID - Account Family ID - Purchase Date - Game Center Ever Enabled - Genre ID - Is Auto Download - Is Factory Install - Item Name	- Label - Artist - Bundle Version - Account Apple ID - Account Downloader ID - Account Purchaser ID - Game Center Enabled - Genre - Messages Extension - Is Purchased - Redownloaded - Item ID - Kind

Attribute	Associated Columns	
	• Launch Prohibited • Rating Rank • Device Based VPP • External Identifier • Variant ID • Data Provenance	• Rating Label • Release Date • Version Bundle ID • Source Application • System Info Provenance
File Application Smart Folder (iOS)	• Name • Item Number • Filename • Last Hit Time • System Info Modified • Is Uploading • Is Downloading • Uploaded • Trashed • In Pinned Folder • Provider Identifier • Content Type Identifier • System Info Provenance	• Label • Folder ID • Summary MD File Type • Created Time • Frequency At Last Hit Date • Is Ubiquitous • Download • Container Pristine • Shared By Current User • Item Identifier • System Info Display Name • User Info • Data Provenance
Mobile Application Permission (iOS)	• Name • Item Number • System Info Service • Application Permission Granted • System Info Provenance	• Label • Bundle ID • Service Type • Associated User • Data Provenance
Mobile Installed Application (iOS)	• Name • Item Number	• Label • User

Attribute	Associated Columns
	• System Info Application Name • System Info Target File Path • Display Version • System Info File Size • System Info Publisher • Container Made Live Time • Container Path • System Info Provenance • Package Name • System Info Install Date • Internal Version • App Store Action • Bundle ID • Container Moved Time • Dumped Location • Data Provenance
Library Notification (iOS)	• Name • Item Number • Bundle Name • Suppress Screen Light Up • Interruption Level • Prevent Notification Dismissal After Default Action • Trigger Time Interval • Present Alert • Ignore Accessibility Disabled Vibration • Sound Repeat • Reply To CurrentUser • App Summary Argument • Mentions Current User • Label • Bundle ID • Created Time • Suppress Sync Dismissal When Removed • UNNotification Center Destination • Ignore Downtime • Play Sound • Tone Alert Type • Use Request Identifier Dismissal Sync • Hide Date • Lock Screen Destination • Icon System Image Name • Trigger Repeat Interval

Attribute	Associated Columns
	• Sound Maximum Duration • Realert Count • Preempt Presented Notification • System Info Notification Type • Should Authenticate Default Action • App Notification Title • Ignore Do Not Disturb • Notification Summary Argument Count • Notify Recipient Anyway • Relevance Score • Footer • Alert Destination • Car Play Destination • Trigger Repeats • Ignore Ringer Switch • Should Hide Time • Context System Image • System Info Provenance • UN Notification Default Destinations • App Notification Identifier • Background Default Action • Media Library Item Identifier • Presentation Options • Context Recipient Count • Notification Title Localization Key • Notification Subordinate Icon • SBS Push Store Category Key • Badge Application Icon • Header • Default Action Key • Critical Alert Sound • Display Actions Inline • Notification Message Localization Key • Schema Version • App Notification Message • Data Provenance

Attribute	Associated Columns	
Stream Notification (iOS)	- Name - Item Number - Timestamp2 - System Info Title - Bundle ID - Additional Identifier - System Info Provenance	- Label - Timestamp1 - Device UUID - Body - Additional Text - Additional Metadata - Data Provenance
Weather Application (iOS)	- Name - Item Number - Latitude (Double) - City Timezone Update Date Key - Last Update Time - Data Provenance	- Label - Longitude (Double) - Time zone - Last Refreshed Time - System Info Provenance
iOS Device Information	- IMEI - Dev Name - Model ID - IMSI - Last iCloud Backup - Find MyiPhone Svc Added Reported Phone# - Bundle ID - Carrier Config Path - Locale ID - iCloud Backup? - App Delete Prohibit?	- Serial# - Display Name - ICCID - Location Services? - Find MyiPhone On? - SIM Phone# - Carrier Config File - Device TZ - Language - iCloud Backup - Encryption Req Last - iCloud Backup TZ - App Install Prohibit?

Attribute	Associated Columns	
	Last System Vers	Provenance
iOS User Shortcut Dictionary	- Word - Locale Category - Path - Description	- User Dict. Freq. - Category - Created
iOS User Word Dictionary	- Word - Locale Category - Path - Description	- User Dict. Freq. - Category - Created
KnowledgeC Activity Level	- Activity - End Date - Provenance - Path	- Start Time - Recorded - Category - Created
KnowledgeC Application Activities	- Provenance - Content Desc - Title - End Date - Category - Created	- Bundle ID - Assoc URL - Start Time - Entry Creation - Path
KnowledgeC Application Focus	- Provenance - Usage In Seconds - Start Time - Entry Creation	- Bundle ID - Generated UUID - End Date - Category
KnowledgeC Application Install States	- Recorded - End Time - App Category - Type	- Start Time - Bundle ID - Application Name

Attribute	Associated Columns	
KnowledgeC Application Intents	- Start Time - Creation Time - Bundle ID - Intent Verb	- End Time - Intent Type - Intent Class
KnowledgeC Application Usage	- Start Time - Entry Creation - Usage In Seconds	- End Time - Bundle ID - Usage In Minutes
KnowledgeC Application Web Usage	- Bundle ID - Start Time - End Date - Category - Created - Data Provenance	- Usage in Seconds - End Time - Entry Creation - Path - Provenance
KnowledgeC Device Lock States	Adj Time	State
KnowledgeC Device Orientation States	- Start Time - Creation Time - Provenance	- End Time - Mobile
KnowledgeC Device Plugged-in States	- Start Time - Is Power Plugged? - Usage in Minutes - Day of Week - Entry Creation - Provenance	- End Time - Usage in Seconds - Title - GMT Offset - Generated UUID
KnowledgeC Do Not Disturb Usage	- Start Time - Creation Time - Usage In Minutes	- End Time - Usage in Seconds - Provenance
KnowledgeC Keybag Lock States	Adj Time	State

Attribute	Associated Columns	
	Provenance	
KnowledgeC Notification Usage	- Bundle ID - Proc Name - End Date - Provenance - Path	- Type - Start Time - Recorded - Category - Created
KnowledgeC Safari History	- Vol UUID - Content Desc - Title - Start Time - Entry Creation - Path - Provenance	- Bundle ID - Source - Generated UUID - End Date - Category - Created
KnowledgeC Screen Backlight States	- Status - End Date - Provenance - Path	- Start Time - Recorded - Category - Created
KnowledgeC Siri UI Usage	- Start Time - Creation Time	- End Time - Bundle ID
Screen Time Application Usage	- Start Time - Usage in Seconds - Device Pickup Count - Given Name	- End Time - Notification Count - Dev Name - Family Name
Screen Time Synced Applications	- Application Name - Dev Name - Family Name	- Device ID - Given Name - Acct Type
Wallet Payment Cards	Card Expiry	Contact Name

Attribute	Associated Columns
	• Full Name • Card Desc • Country Code • Mobile Phone • Card Types(s) • DPAN • Modified • Organization Name • PMT Selected? • Contact Website • Payment Type • Serial# • Creation Time
Wallet Transactions	• Creation Time • Timestamp • Amount • Merchant Industry • Merchant Category • Location Time • Longitude • Horiz Accuracy • Provenance • Modified • Organization Name • Currency Code • Merchant Name • Serial# • Latitude • Elavation • Vert Accuracy • Data Provenance

Communication

Description	This section will contain information relating to general communication items which can be discovered and recovered using the Forensic Tool Kit. These include generic mobile features such as contact lists, SMS/MMS messages, and call information; also found here is information concerning communications and messaging applications. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, names, addresses, and phone numbers may be found under the Contacts attribute.	
Attribute	Associated Columns	
AIM Buddies	• Phone Number • Last Name • Mobile Number • Work Number • Address • Address1 • State • Src • Path • Description	• First Name • Number Type • Home Number • Email • Work Address • Address2 • Zip • Category • Created
AIM Messages	• Sender • Sent • Category • Msg App • Message • Created	• Recipient(s) • Received • Read • # Attachments • Path • Description
Apple Contacts - iOS	• First Name • Last Name • Department • Job Title	• Middle Name • Organization Name • DOB • Creation Time

Attribute	Associated Columns	
	• Modified • Prefix • Note • Middle Name Phonetic • First Name Pronunciation • Last Name Pronunciation • Mobile Number • Email Address • City • Thumbnail Avatar	• Nickname • Suffix • First Name Phonetic • Last Name Phonetic • Middle Name • Work Number • Home Number • Street • FullSize Avatar • Provenance
Discord Messages	• Sender • Sent • Category • Msg App • Message • Created	• Recipient(s) • Received • Read • # Attachments • Path • Description
Facebook Messenger Calls	• Calling Name • Call Number • Call Time • Direction • Country • Path • Description	• Call Type • Call Duration • Src • Category • Created
Facebook Messenger Groups	• Description • Convo Name	• Participants • Topic

Attribute	Associated Columns	
	• Chat Count • Category	• Msg App • Path
Facebook Messenger Messages	• Sender • Sent • Category • Msg App • Message • Created	• Recipient(s) • Received • Read • # Attachments • Path • Description
Facebook Messenger Users Contacted	• Phone Number • Last Name • Mobile Number • Work Number • Address • Address1 • State • Src • Path • Description	• First Name • Number Type • Home Number • Email • Work Address • Address2 • Zip • Category • Created
Google Duo Accounts	• Phone Number • Last Name • Mobile Number • Work Number • Address • Address1 • State • Src • Path • Description	• First Name • Number Type • Home Number • Email • Work Address • Address2 • Zip • Category • Created

Attribute	Associated Columns	
Google Duo Group Calls	- Calling Name - Call Type - Call Duration - Scr - Category - Created	- Call Number - Call Time - Direction - Country - Path - Description
Google Duo Groups	- Description - Convo Name - Chat Count - Category - Created	- Participants - Topic - Msg App - Path
Google Duo Users	- Phone Number - Last Name - Mobile Number - Work Number - Address - Address1 - State - Src - Path - Description	- First Name - Number Type - Home Number - Email - Work Address - Address2 - Zip - Category - Created
Google Hangouts Voice Calls	- Calling Name - Call Type - Call Duration - Country - Path - Description	- Call Number - Call Time - Src - Category - Created
Houseparty Messages	Sender	Recipient(s)

Attribute	Associated Columns	
	• Sent • Category • Msg App • Message • Created	• Received • Read • # Attachments • Path • Description
Houseparty Users	• Phone Number • Last Name • Mobile Number • Work Number • Address • Address1 • State • Src • Path • Description	• First Name • Number Type • Home Number • Email • Work Address • Address2 • Zip • Category • Created
imo Contacts	• Phone Number • Last Name • Mobile Number • Work Number • Address • Address1 • State • Src • Path • Description	• First Name • Number Type • Home Number • Email • Work Address • Address2 • Zip • Category • Created
imo Messages	• Sender • Sent	• Recipient(s) • Received

Attribute	Associated Columns	
	• Category • Msg App • Message • Created	• Read • # Attachments • Path • Description
iOS Call Logs	• Calling Name • Call Type • Call Duration • Country • Path • Description	• Call Number • Call Time • Src • Category • Created
iOS Google Hangouts Cached Images	• File Name • File Size • Provenance	• File Type • External URL • Data Provenance
iOS Google Hangouts Contacts	• Phone Number • Last Name • Mobile Number • Work Number • Address • Address1 • State • Src • Path • Description	• First Name • Number Type • Home Number • Email • Work Address • Address2 • Zip • Category • Created
iOS iMessage/SMS/MMS	• Sender • Sent • Category • Msg App • Message	• Recipient(s) • Received • Read • # Attachments • Path

Attribute	Associated Columns	
	Created	Description
iOS Kik Messenger Attachments	- File Name - File Size	- File Type - External URL
iOS Kik Messenger Messages	- Sender - Sent - Category - Msg App - Message - Created	- Recipient(s) - Received - Read # Attachments - Path - Description
iOS Kik Messenger Users	- Phone Number - Last Name - Mobile Number - Work Number - Address - Address1 - State - Src - Path - Description	- First Name - Number Type - Home Number - Email - Work Address - Address2 - Zip - Category - Created
iOS Snapchat Conversations	- Description - Convo Name - Chat - Msg App - Path Created	- Participants - Topic - Count - Category
iOS Voice Mail	- Sender - Sent - Category - Msg App	- Recipient(s) - Received - Read # Attachments

Attribute	Associated Columns	
	• Message • Created	• Path • Description
Skype Accounts	• Profile Name • Service Type • Country • Category • Created	• Profile • Password • Country Code • Path • Description
Skype Calls	• Calling Name • Call Type • Call Duration • Country • Path • Description	• Call Number • Call Time • Src • Category • Created
Skype Chat Messages	• Sender • Sent • Category • Msg App • Message • Created	• Recipient(s) • Received • Read • # Attachments • Path • Description
Skype Chatsync Messages		
Skype Contacts	• Phone Number • Last Name • Mobile Number • Work Number • Address • Address1 • State • Src	• First Name • Number Type • Home Number • Email • Work Address • Address2 • Zip • Category

Attribute	Associated Columns	
	• Path • Description	• Created
Skype File Transfers	• File Name • File Size • Provenance	• File Type • External URL • Data Provenance
Skype Group Chat	• Description • Convo Name • Chat Count • Category • Created	• Participants • Topic • Msg App • Path
Skype SMS	• Description • SMS From Number • Message • Recipient(s) • Read • Time Received • Src • SMS Country • Path • Created	• SMS Type • SMS To Number • Sender • Incoming • Time Sent • SMS Time Zone of Sender • SMS Hdr. • SMSC Time • Category
Skype Voicemails	• Sender • Sent • Category • Msg App • Message • Created	• Recipient(s) • Received • Read • # Attachments • Path • Description
Slack Channel Messages	• Sender • Time Received	• Time Sent • Category

Attribute	Associated Columns	
	• Read • # Attachments • Path • Description	• Msg App • Message • Created
Slack Channels	• Description • Convo Name • Chat Count • Category • Created	• Participants • Topic • Msg App • Path
Slack Direct Messages	• Sender • Sent • Category • Msg App • Message • Created	• Recipient(s) • Received • Read • # Attachments • Path • Description
Slack Files	• File Name • File Size • Provenance	• File Type • External URL • Data Provenance
Slack Users	• Profile Name • Service Type • Country • Category • Created	• Profile • Password • Country Code • Path • Description
Slack Workspaces	• Description • Convo Name • Chat Count • Category • Created	• Participants • Topic • Msg App • Path

Attribute	Associated Columns	
Snapchat Chat Messages	• Sender • Sent • Category • Msg App • Message • Created	• Recipient(s) • Received • Read • # Attachments • Path • Description
Snapchat Contacts	• Phone Number • Last Name • Mobile Number • Work Number • Address • Address1 • State • Src • Path • Description	• First Name • Number Type • Home Number • Email • Work Address • Address2 • Zip • Category • Created
Snapchat Group Members	• Description • Convo Name • Chat Count • Category • Created	• Participants • Topic • Msg App • Path
Telegram Channel Chats - iOS	• Description • Convo Name • Chat Count • Category • Created	• Participants • Topic • Msg App • Path
Telegram Chats - iOS	• Description • Convo Name	• Participants • Topic

Attribute	Associated Columns	
	• Chat Count • Category • Created	• Msg App • Path
Telegram Messages - iOS	• Sender • Sent • Category • Msg App • Message • Created	• Recipient(s) • Received • Read • # Attachments • Path • Description
Telegram Users - iOS	• Profile Name • Service Type • Country • Category • Created	• Profile • Password • Country Code • Path • Description
WhatsApp for iOS	• Created • Category • Path • L-Size	• Modified • Ext • MD5
iOS WhatsApp Accounts Information	• Phone Number • Last Name • Mobile Number • Work Number • Address • Address1 • State • Src • Path • Description	• First Name • Number Type • Home Number • Email • Work Address • Address2 • Zip • Category • Created

Attribute	Associated Columns	
iOS WhatsApp Chats	• Description • Convo Name • Chat Count • Category • Created	• Participants • Topic • Msg App • Path
iOS WhatsApp Contacts	• Phone Number • Last Name • Mobile Number • Work Number • Address • Address1 • State • Src • Path • Description	• First Name • Number Type • Home Number • Email • Work Address • Address2 • Zip • Category • Created
iOS WhatsApp Groups	• Description • Convo Name • Chat Count • Category • Created	• Participants • Topic • Msg App • Path
iOS WhatsApp Messages	• Sender • Sent • Category • Msg App • Message • Created	• Recipient(s) • Received • Read • # Attachments • Path • Description

Connected Devices

Description	This section will contain information relating to items connectable to the mobile device, which can be discovered and recovered using the Forensic Tool Kit. These include health tracking devices, recording steps and heart rates; also found here is information concerning general Bluetooth and SIM card activity. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found.
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Attribute	Associated Columns
Apple Health Distance	- Start Time - Distance (m) - Duration (s) - DataSource Model - End Time - Distance (ft) - Duration (m)
Apple Health Floors	- DataSource Model - End Time - Start Time - Number of Floors
Apple Health Heart Rate	- DataSource Model - End Time - Provenance - Start Time - Measurement - Data Provenance
Apple Health Steps	- Start Time - Number of Steps - End Time
Bluetooth Devices	- Hw Address - Type - Last Name Updt - Name - Last Svc Updt - Provenance
Seen Bluetooth Devices	- Hw Address - Type - Last Name Updt - Name - Last Svc Updt
SIM Card Activity	- ICCID - Last Update - Home Budget - Number - Slot ID - Roaming Budget

Attribute	Associated Columns
SIM Card ICCID	• ICCID • Slot ID • Roaming Budget • Data Provenance • Number Last Update • Home Budget • Provenance
SIM Card Phone Numbers	• ICCID • Slot ID • Roaming Budget • Data Provenance • Number Last Update • Home Budget • Provenance
SIM Card SMS Messages	• Description • SMS From Number • Message • Recipient(s) • Sent • Time Sent • SMS Time Zone of Sender • SMS Hdr. • SMSC Time • Category • SMS Type • SMS To Number • Sender • Incoming • Read • Time Received • Src • SMS Country • Path • Created

Email and Calendar

Description	This section will contain information relating to general email and calendar items which can be discovered and recovered using the Forensic Tool Kit. These include emails sent and received from providers such as Gmail and Yahoo, as well as Apple Mail. Further found here is calendar information, again from providers such as Google, and those included as built-in applications featured on an Apple mobile device. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, names, email addresses, and message bodies may be found under the Emails attribute.
--------------------	--

Attribute	Associated Columns	
Apple Mail	• Message Row ID • Sender Addr • Account Name • Body File Path • Subject Prefix • Mailbox URL • Snippet • Received Time • Last Viewed Time • Ext UUID	• Sender ID • Sender Comment • Message Body • Subject • Mailbox ID • Thread ID • Sent Time • Created Time • Is Read? • Is Mime Wrapped?
Calendar Events	• Event Summary • Event Description • Event Start Time • Attendees • Repeat Rule • Repeat Until • Priority • Event Status	• Event Category • Location • Event End Time • Repeat Days • Repeat Interval Count • Event Phone Number • All Day

Attribute	Associated Columns
Gmail Emails	• Message Row ID • Sender Addr • Account Name • Body File Path • Subject Prefix • Mailbox URL • Snippet • Received Time • Last Viewed Time • Ext UUID • Sender ID • Sender Comment • Message Body • Subject • Mailbox ID • Thread ID • Sent Time • Created Time • Is Read? • Is Mime Wrapped?
Google Calendar Calendars	• Acct ID • Calendar Name • Has Owner Access? • Internal Data Location • Calendar ID • Calendar Details • To Be Removed? • Provenance
Google Calendar Events	• Event ID • Description • End Time • URL • Event Name • Start Time • Creation Time • Provenance

Encryption and Credentials

Description	This section will contain information relating to Apple KeyChain items which can be discovered and recovered using the Forensic Tool Kit.
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Attribute	Associated Columns	
Apple Keychain Generic Passwords	• Usr Acct • PW/Value • Service • Modified • Path • Provenance	• Display Name • Creation Time • Record# • Category • Created • Data Provenance
Apple Keychain Internet Passwords	• Usr Acct • PW/Value • Service • Modified • Path • Provenance	• Display Name • Creation Time • Record# • Category • Created • Data Provenance
Apple Keychain Saved Credit Cards	• Record# • Modified • Card Expiry • Card Number • Provenance	• Created • Cardholder Name • Card Name • Accessed

Location and Travel

Description	This section will contain information relating to location and travel items which can be discovered and recovered using the Forensic Tool Kit. These include mobile map applications such as Apple Maps, as well as mobility service providers such as Uber. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, locations, directions, and addresses may be found under the Locations attribute.
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Attribute	Associated Columns
Apple Maps Searches	- Creation Time - Latitude - Location Display - Search Qry - Modified - Longitude - Is Nearby Search? - Provenance
Apple Maps Trips	- Ext - Category - L-Size - SHA1 - Created - Path - P-Size - MD5 - SHA256 - Accessed
Cached Locations	- Latitude - Elevation - Speed m/s - Horiz Accuracy - Timestamp - Longitude - Course - MPH - KM/H - Vert Accuracy - Provenance
iOS Google Map Coordinates	- Latitude - Provenance - Longitude - Data Provenance
iOS Maps	- Timestamp - Latitude - Address - Longitude

Attribute	Associated Columns
	- Search Qry - Provenance - Metadata - Data Provenance
iOS Wi-Fi Profiles	- SSID - WiFi Signal Strength - Last Autojoined Time - Last Join Time - BSSID - Hidden Ntwk? - Channel - Data Provenance - Age - Last Updated Time - Pwd Mod Time - Prev Join Time - Autojoined? - Enabled? - Provenance
Parked Car Locations	- Timestamp - Longitude - Loc Uncertainty - Latitude - VIN
Significant Locations	- Entry Time - Creation Time - Latitude - Exit Time - Entry Time (min) - Longitude
Significant Locations Visits	- Entry Time - Creation Time - Latitude - Provenance - Exit Time - Entry Time (min) - Longitude - Data Provenance
Uber Accounts	- Phone Number - Last Name - Mobile Number - Work Number - Address - Address1 - State - Src - First Name - Number Type - Home Number - Email - Work Address - Address2 - Zip - Category

Attribute	Associated Columns	
	- Path - Description	Created
Uber Cached Locations	- Latitude - Elevation - Speed m/s - KM/H - Vert Accuracy - Provenance	- Longitude - Course - MPH - Horiz Accuracy - Timestamp - Data Provenance
Uber Locations		
Uber Profiles	- Profile Name - Service Type - Country - Category - Created	- Profile - Password - Country Code - Path - Description
Uber Rider Payment Details	- Creation Time - Timestamp - Amount - Merchant Industry - Merchant Category - Location Time - Longitude - Horiz Accuracy - Provenance	- Modified - Organization Name - Currency Code - Merchant Name - Serial# - Latitude - Elevation - Vert Accuracy - Data Provenance
Waze Places	- Location Name - Device Long. - Location Country - Location Address - Path	- Device Lat. - City - Location Category - Category - Created

Attribute	Associated Columns	
WiFi Locations	Description	
	Timestamp	Coords
	MAC	Channel
	Info Mask	Speed
	Course	Confidence
	Score	Reach
	Horiz Accuracy	Vert Accuracy
	Latitude	Longitude

Media

Description	This section will contain information relating to media items which can be discovered and recovered using the Forensic Tool Kit. These include generic mobile features such as pictures and videos; also found here is information concerning photo sharing and storage services such as Google Photos. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, capture dates, makes and models of the device used, may be found under the Pictures attribute.
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Attribute	Associated Columns
Audio	- Start Time - Creation Time - Bundle ID - Genre - Usage In Seconds - End Time - Album - Artist - Duration - Usage in Minutes
Carved Video	- Created - L-Size - Capture Date - Model - Longitude - Modified - Category - Make - Latitude - Path
Google Photos Albums	
Google Photos Media	
iOS Snapshots	
Live Photos	- Capture Date - Model - Longitude - Duplicated File - Category - Modified - Make - Latitude - L-Size - Path - Created

Attribute	Associated Columns
Photos Albums	
Photos Media Information	
Pictures	• Capture Date • Model • Longitude • Dupliacted File • Category • Modified • Make • Latitude • L-Size • Path • Created
Videos	• Created • L-Size • Capture Date • Model • Longitude • Modified • Category • Make • Latitude • Path

Mobile Application Attachment

This is applicable for Microsoft Teams, Slack and LinkedIn.

Attribute	Associated Columns	
Attachment	• Label • System Info File Name • System Info File Size • Item Source • System Info Provenance • Name	• Item Number • Summary MD File Type • System Info Target Creation Time • System Info Source • Data Provenance

Media Information

Attribute	Associated Columns	
Business Category Cache	• Name • Item Number • Country ISO • Business Category • Latitude Value • App Update Timestamp • Cache Category • Data Provenance	• Label • MUID • Geo Service Provider • Radius • Longitude Value • Category (Number) • System Info Provenance
For Place List & Sorted List Node	• Name • Item Number • Area • Place Type • Data Provenance	• Label • Dominant Order Type • System Info Name • System Info Provenance
Location Cache	• Name • Item Number • Address Dictionary • Updated Time • Sub Thorough fare • Sub Administrative Area • Ocean • country ISO • Geo Service Provider • Administrative Area • Region Latitude • Latitude Value • Island • System Info Provenance	• Label • Areas Of Interest • Address • Thorough fare • Sub Locality • Region Identifier • Locality(String) • Inland Water • Administrative Area Code • Region Radius • Region Longitude • Longitude Value • Postal Address • Data Provenance

Attribute	Associated Columns	
Postal Address	• Name • Item Number • Formatted Address • Street • Location Country • Locality(String) • System Info Provenance	• Label • State (String) • City • Postal Code • Sub Administrative Area • Country ISO • Data Provenance
Public event query	• Name • Item Number • Latitude Value • Radius • App Update Timestamp • Data Provenance	• Label • Summary MD End Time • Longitude Value • Start Time • System Info Provenance
Reverse Geo Location	• Name • Item Number • Summary Country Code • Version • System Info Provenance	• Label • Home • Address • Geo Service Provider • Data Provenance
Reverse Geo Location	• Name • Item Number • Summary Country Code • Version • System Info Provenance	• Label • Home • Address • Geo Service Provider • Data Provenance

Geo Service

Attribute	Associated Columns	
Geo Service	• Name	• Label
	• Item Number	• Creation Time
	• Count Type	• System Info Application ID
	• Usage String	• Usage
	• System Info Provenance	• Data Provenance

Mobile Phone (iOS)

iCloud Documents

Attribute	Associated Columns	
iCloud Client item	- Name - Item Number - Zone Name - File Length - Modification - System Info Provenance	- Label - System Info File Name - System Info Version - Creation Time - Element Type - Data Provenance
iCloud Server Items	- Name - Item Number - Version Size - Device Name - Birthtime - Last Used Time - System Info Version - Item Type - Data Provenance	- Label - Item Rank - Version Device ID - Quota Bytes Used - Version Modified Time - Filename - Zone Name - System Info Provenance
iCloud Sync Name	- Name - Item Number - Device Name - Data Provenance	- Label - ID - System Info Provenance

Biome

Attribute	Associated Columns	
Audio routing	- Name - Item Number - Connection Status - Device Class - System Info Provenance	- Label - Entry Time - Identifier - Device Name - Data Provenance
Biome Application Launch	- Name - Item Number - Bundle ID - Display Version - System Info Provenance	- Label - Write Time - Associated Launch Reason - Internal Version - Data Provenance
Device Orientation	- Name - Item Number - System Info End Date/Time - System Info GUID - System Info Provenance	- Label - Start Date/Time - System Info Recorded Time - Mobile Orientation - Data Provenance
Screen Backlight	- Name - Item Number - System Info End Date/Time - System Int GUID - System Info Provenance	- Label - Start Date/Time - System Info Recorded Time - Backlight Active - Data Provenance

Device Information

Attribute	Associated Columns	
Mobile Device Activation Record	• Name	• Label
	• Item Number	• LD Activation Version
	• Unbrick	• Device Certificate
	• Account Token Certificate	• Unique Device Certificate
	• Fair Play Key Data	• IMEI 1
	• IMEI 2	• Product Type
	• Unique Device ID	• Serial Number
	• SIM ICCID	• Mobile Equipment Identifier
	• Manufacturing Time	• System Info provenance
	• Data Provenance	

Map Transition

Attribute	Associated Columns	
Mobile Apple Map Transition	• Name	• Label
	• Item Number	• Duration Span
	• Origin Latitude	• Origin Longitude
	• Destination Latitude	• Destination Longitude
	• System Info Target Creation	• Start Date/Time
	• System Info End Date/Time	• System Info Provenance
	• Data Provenance	

Mobile Sharing Daemon

Attribute	Associated Columns
Sharing Daemon	- Name - Item Number - Hash Manager Last Rebuilt Date Key Time - Last Keychain Count Time - Last Sign In Time - Airdrop Random Hash LILIED Key 2 - Airdrop Random Hash LILIID Key 4 - Stream ID - First Name - Validated Email Addresses - Data Provenance - Label - Hash Manager Last updated Date Key Time - Last Key Verification Time - Last All Good Time - Airdrop Random Hash IJIJID Key I - Airdrop Random Hash LILIED Key 3 - Airdrop ID - Apple ID - Last Name - System Info Provenance

Mobile Call

This is applicable for Skype.

Attribute	Associated Columns	
Call	• Name	• Label
	• Item Number	• Calling Name
	• Call Number	• Message Sender
	• Message Recipient(s)	• Call Type
	• Call Time	• Call Duration
	• Direction	• Call Result Status
	• Item Source	• Call Country Code
	• File Type	• Path
	• Created Date	• Item Description

Mobile Contacts

This is applicable for iMessage, Skype, Slack, Microsoft Teams, and LinkedIn.

Attribute	Associated Columns	
Contacts	• Label	• Item Number
	• Phone Number	• First Name
	• Last Name	• Number Type
	• Mobile Number	• Home Number
	• Work Number	• Email
	• Address	• Work Address
	• Address1	• Address2
	• State	• Zip
	• Item Source	• File Type
	• Path	• Created Date
	• Item Description	• Name
	• Avatar Images	• Associate email addresses
	• Associate phone addresses	• Created timestamp

Mobile Chat Conversations

This is applicable for iMessage, Skype, Slack, Microsoft Teams, and LinkedIn.

Attribute	Associated Columns	
Conversations	• Label	• Item Number
	• Item Description	• Chat Participants
	• Conversation Name	• Chat Topic
	• Chat Message Count	• Message Application
	• File Type	• Path

Attribute	Associated Columns
	- Created Date - Name

Mobile Messages

This is applicable for iMessage, Skype, Slack, Microsoft Teams, and LinkedIn.

Attribute	Associated Columns
Messages	- Label - Message Sender - Message Sent Time - File Type - Message Application - Message Body - Created Date - Name - Item Number - Message Recipient(s) - Message Received Time - Message Is Read - Message Attachment Count - Path - Item Description

Operating System

Description	This section will contain information relating to operating system information which can be discovered and recovered using the Forensic Tool Kit. These include features such as Apple accounts information, network usage, and Powerlog information. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, computer name and product version, may be found under the Owner Information attribute
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Attribute	Associated Columns
.DS_Store Records	- Name Len - Struct ID - Summary: Raw Value - Summary Record Value - Name - Struct Type - Summary: Record Type - Provenance
AirDrop Background Activity	- Timestamp - Provenance - Log Msg
AirDrop Discoverability	- Timestamp - Log Msg - Mode
Apple Accounts	- User Name - Account Added - Acct Desc - Acct Cred Type - Last Credential Expiry - Category - Provenance - Acct ID - Parent AcctID - Acct Type - Owning Bundle ID - Path - Created - Data Provenance
Cell Tower Locations	- Timestamp - Mobile Ntwk Code - Loc Area Code - Longitude - Country Code - Cell ID - Latitude - Range

Attribute	Associated Columns	
	Confidence	Provenance
Google Accounts	- Profile Name - Service Type - Country - Category - Created	- Profile - Password - Country Code - Path - Description
Network Usage - Application Data	- Process Last Uded - Last Connected - Proc Name - WiFi Bytes Out - WWAN Bytes Out - Wired Bytes Sent - Mobile Bytes Sent - Provenance	- Process First Used - Bundle ID - WiFi Bytes In - WWAN Bytes In - Wired Bytes Rcvd - Mobile Bytes Rcvd - Type - Data Provenance
Network Usage - Connections	- First Network Attachment Time - Last Network Attachment Time # Conn Attempt - Network Name - Known Good? - Data Provenance	- Timestamp - Network Type # Successful Conn - MAC - Provenance
Owner Information	- Product Name - Product Version - Prod Major Vers - Time Zone - Installed - IOS Major Vers	- Prod C - Prod User Visible Vers - Computer Name - Curr Build # - Country Code - IOS Supported Vers

Attribute	Associated Columns	
	- Category - Created - Data Provenance	- Path - Provenance
PowerLog App Usage	- Start Time - Entry Creation - Usage in Seconds - Provenance	- End Time - Bundle ID - Usage in Minutes - Data Provenance
PowerLog Application State	- Interval Start Time - Bundle ID - Bkg Time (s) - Bkg Location Usage (s) - Data Provenance	- Interval (m) - Screentime (s) - Bkg Aud Played (s) - Provenance
PowerLog Battery Level	- Adj Time - Raw Batt Level - Full Charge - Offset Time - Provenance	- Batt Level - Charging - Orig Time - Time Offset
PowerLog Battery Shutdown	- Adj Time - Raw Batt Level - Full Charge - Offset Time - Provenance	- Batt Level - Charging - Orig Time - Time Offset - Data Provenance
PowerLog Camera State	- Adj Time - Camera Type - Offset Time - Bundle ID	- State - Orig Camera Time - Time Offset - Provenance
PowerLog Device Lock State	- Adj Time - Provenance	- State - Data Provenance

Attribute	Associated Columns	
PowerLog Lightning Cable Status	Adj Time	
PowerLog Process Data Usage	- Adj Time - Bundle ID - Incoming - Wifi Bytes In - Orig Time - Time Offset - Adj End Time - Proc Name - Outgoing - Wifi Bytes Out - Offset Time	
PowerLog Screen Autolock	Adj Time	
PowerLog Timezone Information	- Adj Time - Country Code - GMT Offset - Trigger - Time Offset - Device TZ - Locale ID - TZ DST? - Offset Time	
Siri Message Search Suggestions	- Start Time - Creation Time - Provenance - End Time - Bundle ID - Data Provenance	

Social Networking

Description	This section will contain information relating to general social media items which can be discovered and recovered using the Forensic Tool Kit. These include mobile social media features for a variety of application, such as application users, messages, and contacts. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, profile names and passwords may be found under the Profiles/Users attributes.
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Attribute	Associated Columns
iOS Facebook Friends	- Phone Number - Last Name - Mobile Number - Work Number - Address - Address1 - State - Src - Path - Description - First Name - Number Type - Home Number - Email - Work Address - Address2 - Zip - Category - Created
iOS Facebook Messages	- Sender - Time Sent - Category - Msg App - Message - Created - Recipient(s) - Time Received - Read # Attachments - Path - Description
Instagram Direct Messages	- Sender - Time Sent - Category - Recipient(s) - Time Received - Read

Attribute	Associated Columns	
	• Msg App • Message • Created	• # Attachments • Path • Description
Instagram Group Members	• Description • Convo Name • Chat Count • Category • Created	• Participants • Topic • Msg App • Path
Instagram Media	• File Name • File Size • Provenance	• File Type • External URL • Data Provenance
Instagram Profiles	• Profile Name • Service Type • Country • Category • Created	• Profile • Password • Country Code • Path • Description
iOS Instagram Posts		
iOS Tinder Accounts	• Profile Name • Service Type • Country • Category • Created	• Profile • Password • Country Code • Path • Description
iOS Tinder Messages	• Sender • Time Sent • Category • Msg App • Message • Created	• Recipient(s) • Time Received • Read • # Attachments • Path • Description

Attribute	Associated Columns
iOS Tinder Photos	• File Name • File Size • Provenance • File Type • External URL • Data Provenance

Twitter

Attribute	Associated Columns	
Profile Banner Media	- Label - System Info User Name - DMCA Restricted - Video View Count - Monetizable - System Info Provenance - Name - Item Number - User ID(Number) - Media Type (Number) - Media URL - NFT Avatar - Data Provenance	
User	- Label - Updated Time - Protected User - System Info User Name - Identity Type - Verified - Created Time - System Info URL - Suspended - Followers - Media Count - Pinned Tweet Time - Bio - System Info Provenance - Name - Item Number - Profile Update Time - Full Name - User ID(Number) - Relevance - Blue Verified - Extended Birthday - Location - Following - Followers Normal - Favourites - NFT Avatar - Status Count - Data Provenance	
User Birthday	- Label - System Info User Name - Year Visible - Year - Item Number - User ID(Number) - Day Visible - Month	

Attribute	Associated Columns
	- Day - Data Provenance - System Info Provenance - Name
URL	- Label - System Info User Name - Original Display URL - Expanded URL - Data Provenance - Item Number - User ID(Number) - System Info URL - System Info Provenance - Name
Structured Location	- Label - System Info User Name - Name Value - Full Name - System Info Provenance - Name - Item Number - User ID(Number) - Place ID - Place Type - Data Provenance
Tweets	- Label - System Info User Name - Retweet Count - System Info Language - From User ID - Card Data URL - Updated Time - Auto Translatable - Original Text - Retweeted - Favourites By Loading Account - Status ID - Mention User ID - Symbol - Data Provenance - Item Number - User ID(Number) - System Info Source - Quote Count - Translatable - Reply Count - Date - Possibly Sensitive - Conversation ID - Favourites - Include in Profile Timeline - Retweeted User ID(String) - Hash Tag - System Info Provenance - Name

Attribute	Associated Columns	
Media	- Label - System Info User Name - Media Key - Monetizable - DMCA Restricted - Video View Count - System Info Provenance - Name	- Item Number - User ID(Number) - Media Type (Number) - Media ID - NFT Avatar - Media URL - Data Provenance
Status Information	- Name - Item Number - Modified Date - Extension - MD5 Hash	- Label - Created Date - File Type - Path - Logical Size

Uber

Attribute	Associated Columns	
Cached Location	- Label - Timestamp Date/Time - Reference ID - Sub Title - Longitude Value - Full Address - System Info Category - Address Line1 - Locale - Data Provenance	- Item Number - Uber ID - System Info Title - Latitude Value - Historical (Number) - System Info Name - System Info Provider - Address Line2 - System Info Provenance - Name

Web Related

Description	This section will contain information relating to general web browser items which can be discovered and recovered using the Forensic Tool Kit. These include browser features such as browser history, bookmarks, and downloads; also found here is autofill information, such as credit cards and login details. These are displayed in the Attribute column. In the Associated Columns column, nodes linking to their associated Attribute will be found. For example, searches and corresponding URLs and web page titles may be found under the Search Terms attributes.
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Attribute	Associated Columns
Chrome Archived Keyword Search Terms	- User name - Search Term - Web Page Title - Search Engine - Category - Provenance - Date/Time - URL - Browser - Path - Created
Chrome Archived Web History	- Visit ID - URL - Provenance - This Visit Time - Visit Duration - Data Provenance
Chrome Autofill Profiles	- Browser Origin - Value - Last Use - Provenance - Name - Creation Time - Count - Data Provenance
Chrome Autofill	- Browser Origin - Value - Last Use - Provenance - Name - Creation Time - Count - Data Provenance
Chrome Bookmarks	- Browser - URL - Title Preview Text - Category

Attribute	Associated Columns
	- Description - Creation Time - Modified by Device - Data Provenance - User Defined? - Modified - Provenance
Chrome Cookies	- User Name - Value - Creation Time - Domain - Expires - Category - Provenance - Mane - Last Access - Browser - File Path - Path - Created - Data Provenance
Chrome Current Session	- Browser - Private Window? - Selected Tab Title - Data Provenance - Close Time - Selected Tab Index - Provenance
Chrome Current Tabs	- Browser - Close Time - Tab Index Title - Muted? - Data Provenance - URL - Last Visit - Encrypted? - Provenance
Chrome Downloads	- User Name - URL - File Size - Path - Created - Data Provenance - File Path - Download Time - Browser - Category - Provenance
Chrome FavIcons	- Browser - Last Update - Page URL - Provenance

Attribute	Associated Columns	
	Data Provenance	
Chrome Keyword Search Terms	- User name - Search Term - Web Page Title - Search Engine - Category - Provenance	- Date/Time - URL - Browser - Path - Created
Chrome Last Session	- Browser - Private Window? - Selected Tab Title - Data Provenance	- Close Time - Selected Tab Index - Provenance
Chrome Last Tabs	- Browser - Close Time - Tab Index Title - Muted? - Data Provenance	- URL - Last Visit - Encrypted? - Provenance
Chrome Logins	- Login Row ID - Action URL - Username Value	- Origin URL - Username Element - Password Element
Chrome Saved Credit Cards	- Card Type(s) - Expiration Year - Name on Card - Status - Data Provenance	- Expiration Month - Last 4 - Issuer - Provenance
Chrome Shortcuts	- Shortcut - Text - URL - Number Hits	- Row ID - Fill In Edit - Last Access

Attribute	Associated Columns	
Chrome Tab History	- ID - URL - Original Request URL - Transition Type	- Index Entry - Title - Referrer URL
Chrome Top Sites	- Browser - Title - Category - Provenance	- Builtin? - URL - Description - Data Provenance
Chrome Web History	- Visit ID - URL - Visit Count - Data Provenance	- Last Visit - Title - Provenance
Chrome Web Visits	- Visit Date - URL - Referring URL	- Visit Count - Title
Edge Chromium Bookmarks	- Browser - URL - Description - Creation Time - Modified by Device	- Title Preview Text - Category - User Defined? - Modified
Edge Chromium FavIcons	- Browser - Last Update - Data Provenance	- Page URL - Provenance
Edge Chromium Logins	- Login Row ID - Action URL - Username Value	- Origin URL - Username Element - Password Element
Edge Chromium Web History	Visit ID	Last Visit

Attribute	Associated Columns	
	• URL • Visit Count	• Title • Provenance
iOS Safari Recent Search Terms	• User Name • Search Term • Web Page Title • Search Engine • Category • Provenance	• Date/Time • URL • Browser • Path • Created • Data Provenance
Malware/Phishing URLs	• Name • URL • Category • Description • Metadata • Category • Provenance	• Last Visit • URL Title • Sub Category • Activity • Path • Created • Data Provenance
Safari Bookmarks	• Browser • Preview Test URL • Description • Creation Time • Modified By Device	• Title • Category • User Defined? • Modified • Provenance
Safari Downloads	• User Name • URL • File Size • Path • Created	• File Path • Download Time • Browser • Category
Safari History	• Start Time • Title	• End Time • URL

Attribute	Associated Columns
	• Bundle ID • GMT Offset • Generated UUID • Day of Week • Entry Creation • Provenance
Safari iCloud Devices	• Browser • Dev Name • Modified • Creation Time • Device UUID • Provenance

7 Decryption Support

FTK can decrypt the following encryption types:

- Bitlocker
- iOS Backup
- Android Backup
- Safeguard (Sophos)
- McAfee Safeboot
- PGPDisk
- Guardian Edge
- AFF
- PointSec
- FileVault (Apple)
- Geli (Linux, BSD)
- TrueCrypt and compatible: VeraCrypt and DiskEncryptor
- WinMagic, aka SecureDoc
- Dell full disk encryption
- APFS
- McAfee MDE
- Vera Crypt
- Encrypted iOS File System Cellebrite

8 Cloud Exports

Fitbit

Applicable for the following Sub-Nodes:

- Activity Information
- Daily Heart rate information
- Device information
- Exercise
 - Exercise Session
 - Exercise Events
- Sleep Information
 - Sleep level
 - Sleep log
- User Profile

The following are the list of default column set:

- | | | |
|----------------|-----------------|-----------------|
| • Name | • Label | • Item |
| • Created Date | • Modified Date | • Category |
| • Extension | • Path | • MD5 |
| • L-size | • Object Name | • Labels |
| • Object ID | • Created Date | • Last Modified |
| • File Type | • Physical Size | • Extension |
| • Logical Size | • Path | • MD5 Hash |
| • SHA1 Hash | • SHA256 Value | • |

Google Files

The following are the list of default column set:

- Item
- File Size In Bytes
- Duration
- Root Path
- Mime Type Str
- Parent Folder Name
- Data Provenance
- Name
- Pixel Width
- Track Number
- Root Relative File Path
- Uri
- Is Hidden
- Modified Time
- Media Store Id
- Pixel Height:
- Iso Speed
- File Name
- Title
- Provenance

Android Turbo (Device Health Services)

Mobile Turbo Application Launch Event (Android)

The following are the list of default column set:

- Object Id
- Package Name
- Provenance
- Object Name
- Launch Time
- Data Provenance

Mobile Turbo Battery Event (Android)

The following are the list of default column set:

- Object Id
- Battery Level Value
- Time Zone
- Data Provenance
- Object Name
- Charge Type
- Provenance
- Timestamp Time

Google Cast

The following are the list of default column set:

- Name
- Object ID
- Device ID
- Model name
- Device Instance Name
- Capabilities
- Last Discovered time
- Data Provenance
- label
- Object Name
- Device Version
- Device service ID
- Supported Criteria
- RCN Enabled Status
- Last Discovered BLE time
- item
- Label
- Friendly name
- Receiver Metrics ID
- Cloud Device ID
- Last Published time
- Provenance

Mobile Google Playstore Installed Application (Android)

The following are the list of default column set:

- Object Id
- Application Name
- Install Client ID
- Isauto Update Enabled
- Data Provenance
- First Download Time
- Object Name
- App Install Reason
- Title
- Desired Version
- Installed App Purchased Date
- Install App Install Timestamp
- Associate Account
- Download Bytes Total
- Last Notify Version
- Provenance
- Data Delivery Time

Contact Exterro

If you have any questions, please refer to this document, or any other related materials provided to you by Exterro. For usage questions, please check with your organization's internal application administrator. Alternatively, you may contact your Exterro Training Manager or other Exterro account contact directly.

For technical difficulties, support is available through support@exterro.com.

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